

Evaluating isolation and contact tracing for control of influenza outbreaks with pandemic potential: a modelling study

Joshua W. Lambert, Sebastian Funk, Carl A.B. Pearson, and Adam J. Kucharski

Abstract

Introduction

Outbreaks of severe respiratory infections pose a major public health threat. The COVID-19 pandemic resulted in over 7 million reported deaths (WHO COVID-19 Dashboard), while the 1918 influenza pandemic caused an estimated ~ 50 million deaths (Johnson and Mueller, 2002). Influenza subtypes to which populations have low levels of pre-existing immunity are a particular concern as potential pandemic candidates. This includes influenza A/H5N1 (Ward et al., 2024) and A/H7N9 (Tanner et al., 2015), as well as A/H1N1pdm when it caused a pandemic in 2009 (Fraser et al., 2009). Recent human cases of avian influenza A/H5N1, alongside transmission, further illustrate the ongoing potential for viruses to infect new hosts, creating the potential for further adaption (Antia et al., 2003; Garg et al., 2025; Peacock et al., 2025). It is therefore crucial to understand the effectiveness and feasibility of different intervention strategies in the face of such infections (Fraser et al., 2004; Kucharski, 2024).

Early in a novel outbreak, pharmaceutical options such as treatments and vaccines may not yet be available, leading to reliance on non-pharmaceutical interventions (NPIs) to reduce infection and disease. These interventions include individual-level targeted measures such as isolation and contact tracing, to population-level policies such as venue closures or restrictions on social mixing (Haug et al., 2020; Sharma et al., 2021). The success of such interventions in curtailing transmission before it becomes sustained depends on several factors, including: the type of intervention, timing of introduction, and the intervention’s effectiveness (Longini et al., 2005). For highly transmissible pathogens with a short generation time (e.g. pandemic influenza), incidence of new infections can grow quickly, outpacing a response. Moreover, pathogens such as influenza, which have a proportion of onward transmission prior to symptom onset and have asymptomatic (subclinical) cases, are difficult to control via symptom-based isolation of cases, potentially in combination with tracing and quarantining of their contacts (Fraser et al., 2004).

Several pre-COVID-19 studies concluded that influenza viruses cannot be controlled by isolating individuals at symptom onset and tracing their contacts, because the rate at which infected individuals become infectious and transmit is faster than the latency in the process of isolating contacts (Fraser et al., 2004; Klinkenberg et al., 2006). However, with the recent human infections of avian influenza (A/H5N1), mostly among poultry and cattle farmers (Garg et al., 2025) and the longer estimates of incubation period and serial interval from historical A/H5N1 data (Ward et al., 2024) it may be that isolation and contact tracing is able to control influenza subtypes with pandemic potential.

In this study we evaluate the effectiveness of controlling outbreaks that exhibit influenza-like epidemiological characteristics using a mathematical model of isolation and contact tracing. In particular, we investigate the assumption that influenza will inevitably outpace our ability to trace and isolate sufficiently effectively for control, and whether this assumption holds for all pathogen subtypes given emerging estimates of incubation period and serial interval (Ward et al., 2024). Finally, we quantify the speed and effectiveness of isolation for containment under different response delays across epidemic scenarios, assessing the impact of reducing the delay to isolation on outbreak control with the use of rapid testing, and the speed-sensitivity trade-off for tests.

Methods

Transmission model

We used a branching process transmission model to assess the potential effectiveness of isolation and contact tracing to control an outbreak of pandemic-potential influenza, implemented in the R package `{ringbp}` (Figure 1) (Hellewell et al., 2025). In the model, each infectious individual generates secondary cases from a negative binomial distribution with a given basic reproduction number (R_0) and dispersion parameter (k). The number of secondary cases sampled for each infector is independent and identically distributed. Onwards transmission from infectious individuals can be reduced in the following

ways: 1) symptomatic individuals isolate, 2) contacts of symptomatic individuals are traced and isolate if they become symptomatic, 3) contacts of symptomatic individuals are traced and quarantined immediately. Symptomatic individuals that have been traced (i.e. infectees) are isolated at whichever of two events occurs first, either 1) after the onset-to-isolation delay after symptom onset, or 2) when their infector is isolated (if quarantined or symptomatic) or when they become symptomatic if the infector is already isolating.

We included several mechanisms by which isolation and contact tracing can be imperfect at reducing onward disease transmission. The proportion of contacts ascertained by contact tracing can range between 0% (i.e. no contacts are traced) to 100% (i.e. every contact of every symptomatic case is traced). Because the model focuses on NPIs reducing onwards transmission, identifying 50% of cases and preventing 100% of their onwards transmission via contact tracing is equivalent to identifying 100% of cases and preventing 50% of their onwards transmission. Subsequent symptomatic cases that have not been contact traced are only isolated after the symptom onset-to-isolation delay. Additionally, we assumed infections have an independent probability of being asymptomatic (or paucisymptomatic); these infections are not identified and therefore their contacts are not traced. Unlike a model with detailed transmission in subpopulations which may not be known early in a novel outbreak, the advantage of this approach is that it enables estimation of the overall effect of multiple interventions on population transmission with relatively few parameters.

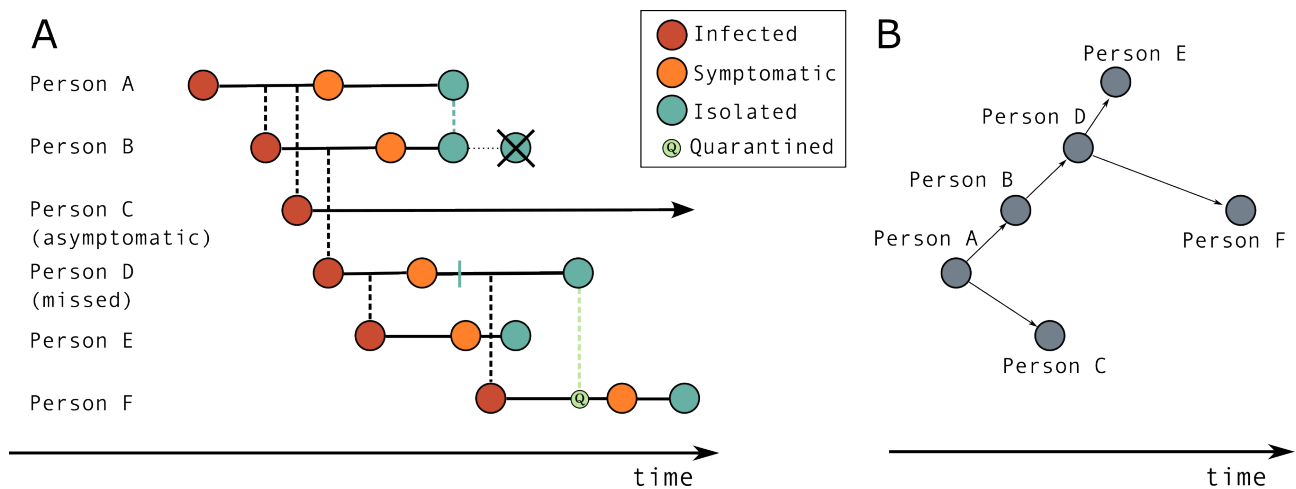


Figure 1: Model schematic of the branching process with isolation, contact tracing and quarantine implemented in the `{ringbp}` R package. (A) The transmission chain from the index case (Person A), infection events are shown by black vertical dashed lines. Individuals can be infected (red circle), become symptomatic (orange circle), and isolated (blue circle). Quarantine is optional in the model, assuming quarantine is active contacts are quarantined when the infector is isolated independent of the contacts symptom status (green circle with Q). The time at which an infector's isolation triggers the isolation of a symptomatic case is shown by the blue vertical dashed line (Person A to Person B). The isolation times that would have been triggered after a delay from symptom onset, but are after isolation was triggered by contact tracing is shown by a blue circle with a cross and connected by a dotted line (Person B). Person C is an asymptomatic individual (asymptomatic cases can infect others). Person D is missed by contact tracing (contacts are traced with a probability ρ , and thus missed at $1 - \rho$). The time Person D would have isolated if ascertained is shown by the vertical blue mark. Assuming quarantine is active, Person F would be isolated before symptom onset at the time of the infector's (Person D) isolation time. If quarantine is not active, then Person F would be isolated after a delay from symptom onset, as they had yet to show symptoms when the infector was isolated. (B) A branching process diagram showing who-infected-whom.

Pathogen epidemiological parameters

In this study we evaluate the feasibility of controlling outbreaks of A/H5N1-, A/H1N1-, and A/H7N9-like infections. The generation time is modelled using the incubation period and the proportion of presymptomatic transmission. A skew-normal distribution is parameterised with a case's symptom onset time and the proportion of transmission prior to symptom onset, and then the distribution is sampled to give the infectee's infection time. This simulates a correlated structure between the two delay distributions to generate realistic delays between exposure, symptom onset and onward transmission (Figure 2A) (Hellewell et al., 2020; Linton et al., 2022). We extracted previously estimated incubation periods from the literature for each subtype. Nishiura and Inaba (2011) report the best fitting Weibull shape and scale parameters for H1N1 to be 1.77 and 1.86, respectively (Figure 2B). Cowling et al. (2013) report the incubation period for H5N1 and H7N9, both Weibull distributed. For H5N1 the Weibull distribution has a mean of 3.3 days and standard deviation (SD) of 1.5 days, for H7N9 the Weibull distribution has a mean of 3.1 days and SD 1.4 days (Cowling et al., 2013) (Figure 2B). Comparatively, the incubation period for H1N1 is shorter than the other two subtypes (H5N1 and H7N9), which are similar, with H5N1

having a slightly longer median (Figure 2B).

We considered four basic reproduction number (R_0) values when simulating transmission in the community: 1.1, 1.5, 2.5 and 3.5 (Figure 3). These are based on similar values to previously estimated R_0 values for influenza outbreaks (Ferguson et al., 2006), including A/H1N1pdm (Fraser et al., 2009; Lessler et al., 2009), or they represent inflated values of current estimates of A/H5N1 and A/H7N9 subtypes to evaluate the effectiveness of isolation and contact tracing in the event that pathogen evolution enhances the human-to-human transmissibility (e.g. genetic reassortment (Peacock et al., 2025) or mutations in receptor binding preference (Lin et al., 2024)). We modelled two scenarios of isolation effectiveness; the first, perfect isolation where isolated cases are assumed to not infect anyone ($R_0^{\text{iso}} = 0$), the second, an imperfect isolation where isolated cases can still infect others but at a reduced reproduction number compared to community transmission ($R_0^{\text{iso}} = 0.5$). The imperfect isolation scenario captures when people isolate at home with others in the household that can be infected, and subsequently go into the community to sustain transmission.

Influenza A viruses have shown moderate individual-level variation in transmission (Fraser et al., 2009; He et al., 2020; Ward et al., 2024). The dispersion parameter (k) for the negative binomial offspring distributions was set to either 0.8 or 5.0, to model moderate transmission heterogeneity ($k = 0.8$), and a relatively homogeneous transmission scenario ($k = 5$); to investigate its role in outbreak containment with individual-level interventions.

We simulated three presymptomatic transmission scenarios: 1%, 15%, and 30% of transmission occurring prior to symptom onset. We assumed that cases can be asymptomatic and infect others, but the majority of cases develop symptoms at some point while infectious. Asymptomatic cases remain in the community throughout their infectious period. We model two transmission modes: airborne transmission and droplet transmission (Tellier et al., 2019). For airborne transmission, asymptomatic cases are assumed to have the same infectiousness as symptomatic cases, so are modelled with the same reproduction number and transmission heterogeneity. Droplet transmission assumes cases need to expel droplets (e.g. via coughing and sneezing), and thus asymptomatics will infect fewer secondary cases. The droplet transmission scenario set the basic reproduction number of asymptomatic cases (R_0^{asym}) to 0.5 and used the same dispersion parameter as symptomatic cases in the community. For each transmission mode we ran scenarios with 10% of cases being asymptomatic, and a more extreme scenario where 30% of cases never develop symptoms and thus are not isolated and secondary cases are missed by contact tracing. We also include a scenario in which all cases develop symptoms (0% asymptomatic cases). These proportions of presymptomatic transmission and asymptomatic fraction cover a broad range of possible pandemic influenza possibilities (Figure 3) (Lessler et al., 2009; Xu et al., 2013; Montgomery et al., 2024; Garg et al., 2025).

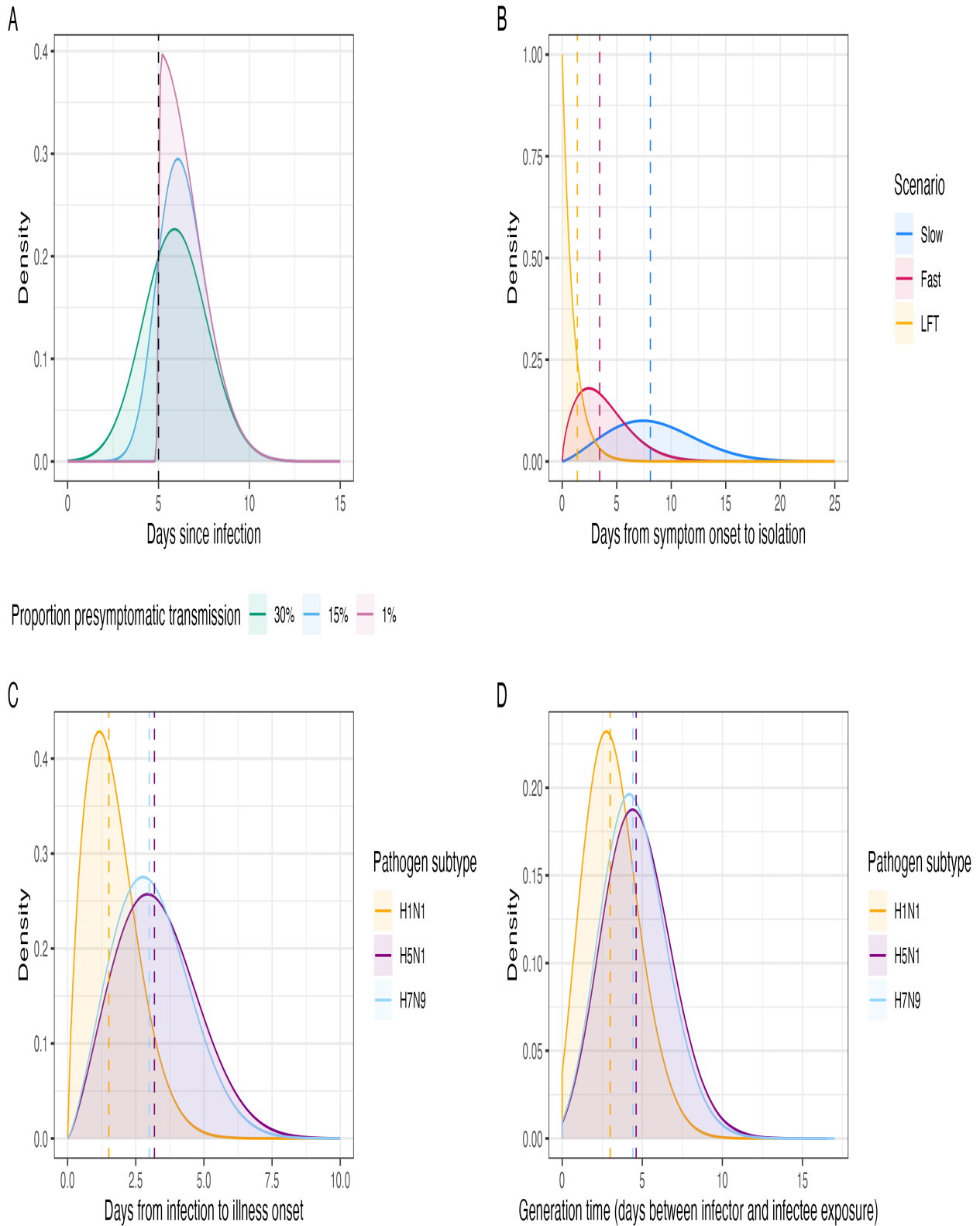


Figure 2: (A) Proportion of presymptomatic transmission determined by the incubation period, here assumed to be 5 days (dashed vertical line), controlled by a skew-normal distribution. Three skew-normal distribution probability density functions are plotted with different skew parameters, 0.73, 1.96, 31.82, which correspond to 30%, 15% and 1% presymptomatic transmission. (B) Incubation period for the influenza subtypes: A/H1N1, A/H5N1 and A/H7N9. The vertical dashed lines are the median incubation period: A/H1N1 is 1.5 days (yellow dashed line), A/H5N1 is 3.2 days (purple dashed line), and A/H7N9 is 3.0 days (blue dashed line). (C) Three assumed onset-to-isolation delays used in the simulation experiment. The *Slow* scenario is based on the onset-to-isolation for the SARS-CoV-2 response in Wuhan, China (Li et al., 2020), the median onset-to-isolation is 8.1 days (blue dashed line). The *Fast* scenario is based on the onset-to-isolation for SARS-CoV-1 response in Hong Kong (Donnelly et al., 2003), the median onset-to-isolation time is 3.4 days (red dashed line). The lateral flow test (LFT) scenario is an assumed distributed for rapid self-administered home testing upon isolation, the median delay from symptom onset to isolation in this scenario is 1.4 days (orange dashed line). (A) are individual-level distributions, (B) & (C) are population-level distributions.

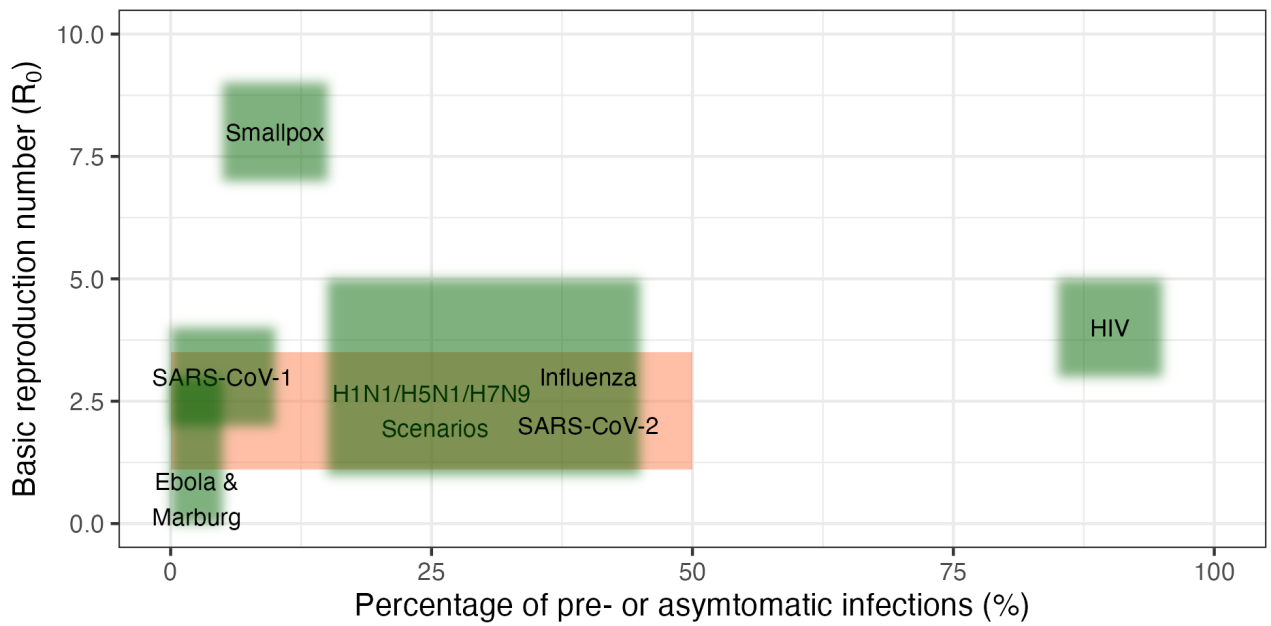


Figure 3: The pathogen characteristics, in terms of transmissibility (R_0) and proportion of disease transmission that occurs before the onset of symptoms (including cases that are entirely asymptomatic). The figure includes: Influenza, SARS-CoV-1, SARS-CoV-2, Smallpox, Ebola, Marburg, and HIV shown by green shaded areas. The size of the shaded area reflects the differences in pathogen characteristics, both between variants and outbreaks, the blurred edges reflects uncertainty in parameter estimates. The parameter space of transmissibility and pre- and asymptomatic transmission explored in this paper is shown in the orange box. Based on Figure 2 in Fraser et al. (2004) but with updated values of R_0 and pre- and asymptomatic transmission extracted from the literature.

Outbreak response scenarios

The onset-to-isolation delay was parameterised under three response scenarios. A fast Polymerase Chain Reaction (PCR) response, a slow PCR response, and a lateral flow rapid test (LFT) response. The time delay from symptom onset to isolating varies between countries and pathogen, as well as during an outbreak (Donnelly et al., 2003; Ali et al., 2022). The slow PCR response was modelled with a mean of 5 days and a SD of 2 days, while the fast PCR response was modelled with a mean of 3 days and a SD of 2 days (Figure 2C). Both fast and slow responses were parameterised as Weibull distributions (fast: $\lambda = 1.53, k = 3.33$; slow: $\lambda = 2.70, k = 5.62$). Our estimates of fast and slow PCR response delays were informed by multiple outbreaks across several regions (see supplementary methods for onset-to-isolation delays reported in the literature). The third onset-to-isolation scenario was based on a situation where LFTs are provided to all susceptible individuals in the population, and these people are freely able to self-test if they become symptomatic. The LFT response was parameterised as an exponential distribution with mean of 1 day ($\lambda = 1$), where most individuals will self-test very shortly after they become symptomatic (approximately 63% of cases will self-test within the first day of symptom onset), with a few people waiting a 2 or more days before testing (less than 0.1% take longer than one week to take a self-test) (Figure 2C). We assume that PCR and LFT sensitivity is 100% accurate so all symptomatic individuals will receive a positive test result.

To assess the feasibility of outbreak containment under different proportions of contacts traced and isolated we varied the proportion of contacts ascertained between 0% (i.e. no contacts traced) and 100% (i.e. all contacts traced) at intervals of 20%. When contact ascertainment is zero, isolation is the only NPI reducing transmission. We compared scenarios where traced contacts only isolate once they become symptomatic (contact trace and isolate), against traced contacts quarantining as soon as the infector isolates (contact trace and quarantine). To test scenarios where an outbreak starts with a different number of initial infections, we simulated with either 5, 20 or 40 initial independent cases. It is assumed that each initial infection seeds an independent outbreak, for example, multiple importations of cases in different parts of the country.

Defining outbreak control

We define outbreak control as no new cases after 12 weeks from the start of an outbreak. The rationale for this criteria is that some outbreaks that do not take off and become sustained may still have cases for the first few weeks of an outbreak, therefore the window for control needs to be enough time after the seeding of an outbreak to be sure it is contained (Hellewell et al., 2020). Due to the exponential dynamics of outbreaks and no depletion of susceptibles in our model we implement a maximum number of cases in each simulation. When this cumulative case limit is reached, it is judged to be a sustained epidemic, not controllable within 12 weeks. We set this limit to 5,000 cases. The percentage of outbreaks controlled was calculated across 1,000 simulation replicates for each scenario.

Testing speed-sensitivity trade-off

Thus far, all scenarios have assumed 100% test sensitivity (i.e. all tests of infected cases return a positive test result). Empirically, PCR tests have a near 100% test sensitivity, but sensitivity is lower for LFTs. LFT sensitivity is around 54% for influenza and 78.8% for COVID-19 (Merckx et al., 2017; Peto et al., 2021). We estimated the minimum LFT test sensitivity required for LFT-based response to match or exceed the fast and slow PCR responses with an assumed PCR sensitivity of 100%. For this analysis we only model the epidemiological characteristics of the H5N1 influenza subtype, assuming a community R_0 of 2.5, 15% of transmission before symptom onset, and 10% of cases are asymptomatic. The proportion of contacts ascertained in tracing was set to 80% and outbreaks seeded with 5 initial cases. These parameters represent midpoint values from the broader scenario analysis.

We quantified the performance of isolation and contact tracing by the proportion of outbreaks controlled within 12 weeks, as described above. To find the minimum LFT test sensitivity whereby the proportion of outbreaks controlled equals slow and fast PCR testing responses, we defined an objective function which simulates the LFT scenario with the test sensitivity as the only free parameter and calculates the absolute difference in control proportion relative to the PCR baseline. The objective function was optimised and all proportion of outbreaks controlled for PCR or LFT scenarios were calculated for 1,000 simulation replicates.

Results

The reproduction number, contact tracing ascertainment and speed of isolation all affect the ability to control a pandemic potential influenza outbreak (Figure 4). Evaluating a representative pandemic influenza scenario (20 initial cases, 15% presymptomatic transmission, 10% asymptomatic), we find isolation alone can control between 25%-50% of epidemics with an R_0 of 1.1 with a slow onset-to-isolation response (mean 5 days); while an isolation-only fast (mean 3 days) or LFT (mean 1 day) response almost always controls outbreaks when $R_0 = 1.1$ (Figure 4). When $R_0 = 1.5$, over 80% of contacts need to be traced to control at least 75% of outbreaks for a slow onset-to-isolation, whereas 40% of contacts traced controls 75% of outbreaks under a fast isolation delay (Figure 4A,B). Isolating after self-testing with an LFT controls epidemics with

$R_0 \leq 1.5$ without tracing contacts (Figure 4C). High transmissibility scenarios ($R_0 \geq 2.5$) with 100% of contacts traced are uncontrolled in almost all slow PCR responses and more than half of fast PCR responses. Comprehensive contact tracing (100%) with LFTs controls almost all outbreaks with $R_0 = 2.5$, but is unlikely to contain an outbreak with $R_0 = 3.5$ (Figure 4C).

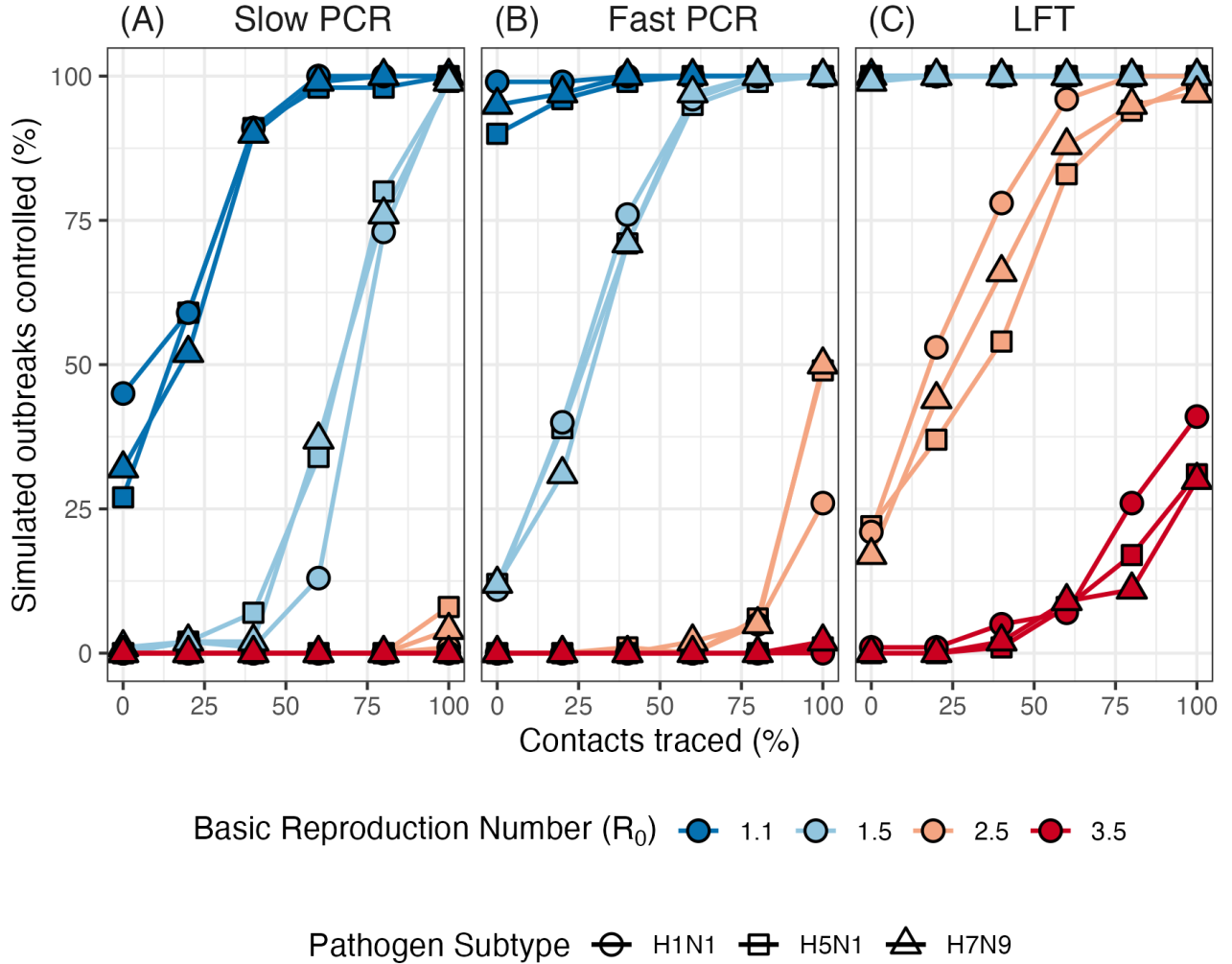


Figure 4: The percentage of outbreaks controlled across percentage of contacts traced. Outbreaks vary by basic reproduction number ($R_0 = 1.1, 1.5, 2.5, 3.5$), and influenza subtype (H1N1, H5N1, H7N9). The plot is faceted by onset-to-isolation response: (A) slow PCR, (B) fast PCR, (C) LFT. This outbreak simulation has 20 initial cases, 15% presymptomatic transmission, 10% of cases are asymptomatic and contacts are isolated when they become symptomatic (i.e. not quarantined).

For the three influenza subtypes analysed, the influence of subtype on the likelihood of control for a given set of NPIs is minimal. There is not a strong pattern for the variance in outbreak control between influenza subtypes across scenarios; the few scenarios with higher variance between subtypes are at high percentages of contacts traced ($\geq 80\%$) and for high transmissibility scenarios (Figure S1).

The probability of outbreak control is greater with perfect isolation ($R_0^{\text{iso}} = 0$), compared to leaky isolation ($R_0^{\text{iso}} = 0.5$) (Figure S2). Most notably low transmissibility scenarios ($R_0 \leq 1.5$) require lower contact traced coverage to control at least 75% of outbreaks. Imperfect isolation reduces the ability to control a $R_0 = 2.5$ scenarios to near zero for LFT responses, whereas perfect isolation can control at least 50% of outbreaks when at least 80% of contacts are traced (Figure S2). Increased individual-level transmission heterogeneity has a minimal, but positive effect to contain the outbreak across subtypes, isolation speed, and contact tracing coverage (Figure S3).

Hitherto all results refer to airborne transmission scenarios which assumes equal transmissibility from symptomatic and asymptomatic cases in the community ($R_0 = R_0^{\text{asym}}$). The droplet transmission scenario (assuming asymptomatic case transmission is subcritical, $R_0^{\text{asym}} = 0.5$), shows broadly the same pattern of containment probability across influenza sub-

types, transmissibility, and isolation responses (Figure S4 and S5). The prominent difference between airborne and droplet scenarios is the latter shows a greater proportion of outbreaks controlled at higher proportions of asymptomatic cases (Figure S4). Whereas in the airborne scenario asymptomatic cases evade being isolated and their contacts traced, hampering containment, in the droplet scenario asymptomatic cases, on average, do not continue the outbreak ($R_0^{\text{asym}} < 1$). All results hereafter refer to airborne scenarios with equal transmissibility between symptomatic and asymptomatic cases.

Increased presymptomatic transmission and asymptomatic cases decreases epidemic containment (Figure S5 and S6). When 30% of transmission occurs before symptom onset, in high transmissibility scenarios ($R_0 \geq 2.5$) more than half of outbreaks become sustained epidemics for all onset-to-isolation delays (Figure S6). When only 1% of transmission is presymptomatic and contact tracing ascertains all contacts, outbreaks with $R_0 = 3.5$ are contained in $>75\%$ of outbreaks with LFT testing, and in $\sim 25\%$ of outbreaks with a fast isolation response (Figure S6). When all cases develop symptoms, isolation and contact tracing all contacts can control all outbreak scenarios, with the sole exception being A/H1N1 with $R_0 = 3.5$ and a slow onset-to-isolation (Figure S5). When 10% of cases are asymptomatic the ability to contain highly transmissible influenza ($R_0 \geq 2.5$) dramatically decreases, and decreases further when 30% of cases are asymptomatic or paucisymptomatic, even with comprehensive contact tracing and isolation upon LFT self-testing, less than a quarter of outbreaks are contained (Figure S5).

The more cases that initially seed an outbreak with $R_0 \geq 1$, the less chance that it will stochastically go extinct and we find that the higher the number of initial influenza cases reduces the efficacy of control-by-isolation (Figure 5). Focusing on the fast PCR response, isolation alone can control most outbreaks with $R_0 = 1.1$ even when seeded with 40 initial cases (Figure 5). High transmissibility ($R_0 \geq 2.5$) scenarios that are seeded with at least 20 cases cannot be controlled in the majority of outbreaks regardless of subtype or contacts ascertained (Figure 5).

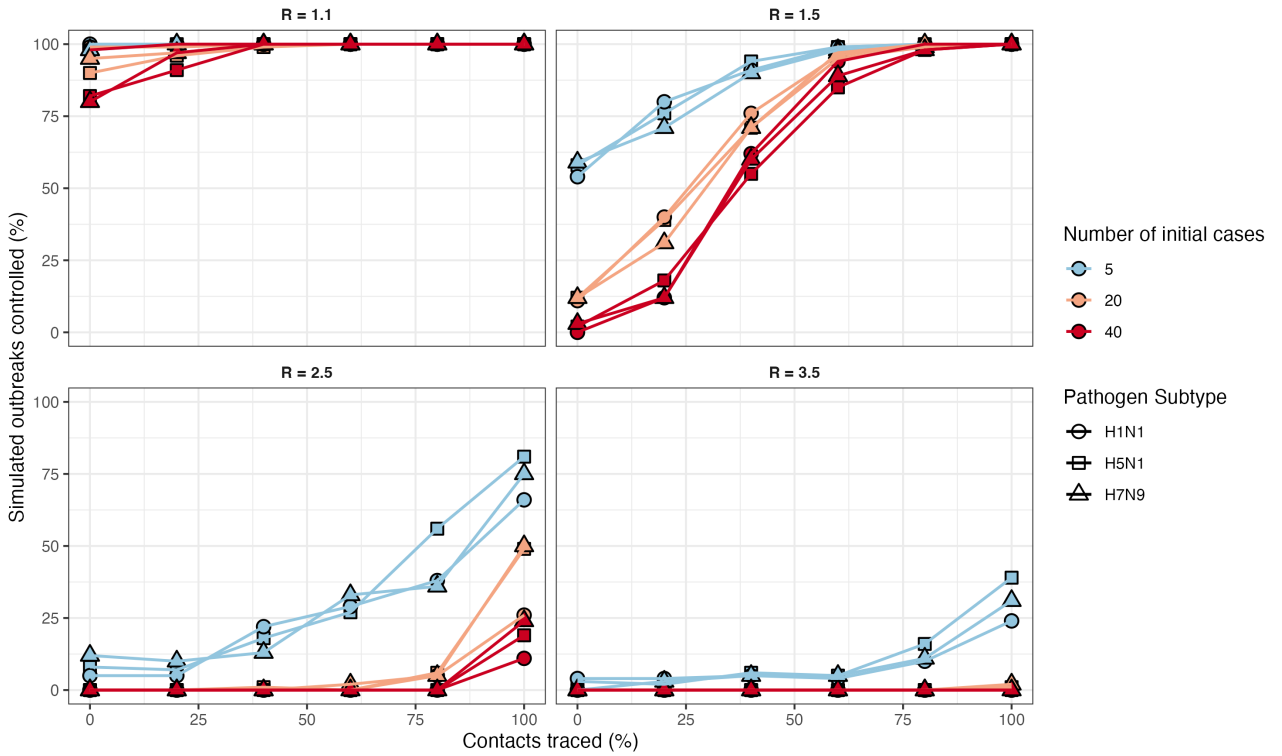


Figure 5: The percentage of outbreaks controlled across percentage of contacts traced. The plot is faceted by basic reproduction number ($R_0 = 1.1, 1.5, 2.5, 3.5$). The number of initial cases seeding the outbreak is varied ($N = 5, 20, 40$). This is a baseline scenario where the onset-to-isolation delay is assumed to be the fast PCR response (mean 5 days, SD 2 days), the proportion of presymptomatic transmission is 15%, 10% of cases are asymptomatic and contacts are isolated when they become symptomatic (i.e. not quarantined).

In the acute phase of outbreaks there can be thousands of weekly cases, especially for high transmissibility scenarios ($R_0 \geq 2.5$) (Figure S7). When outbreaks are contained, the outbreak size is mostly below 1,000 cases (Figure S8). Thus, outbreak control has the dual benefit of reducing the peak weekly incidence as well as the outbreak size.

The control of influenza outbreaks with a $R_0 > 1$ require interventions to bring R_t below 1 for transmission to eventually cease. Isolation of symptomatic cases alone reduces R_t (Figure 6). When the delay between symptom onset and isolation

is on average 5 days (slow PCR response), the reduction in R_t can be minimal, whereas if isolation is rapid after symptom onset, around 1 day (LFT self-testing response), isolation without contact tracing or quarantine brings R_t below 1 when $R_0 < 3.5$ (Figure 6). Contact tracing can further suppress transmission, with 100% contact ascertainment bringing an outbreak under control ($R_t < 1$) for all influenza subtypes and isolation responses for $R_0 < 2.5$ (Figure 6). Quarantine has a negligible impact on the reducing disease transmission for the scenarios we investigated (Figure 6 and S9). The effectiveness of tracing and isolating contacts is reduced when more cases transmit before symptomatic or are asymptomatic (Figure 6 and S9).

The test sensitivity for LFTs needs to exceed 76.4% for it to outperform the outbreak containment of the fast PCR response. An LFT test sensitivity of 7.7% or greater produces an outbreak response that has at least the same containment probability as the slow PCR response.

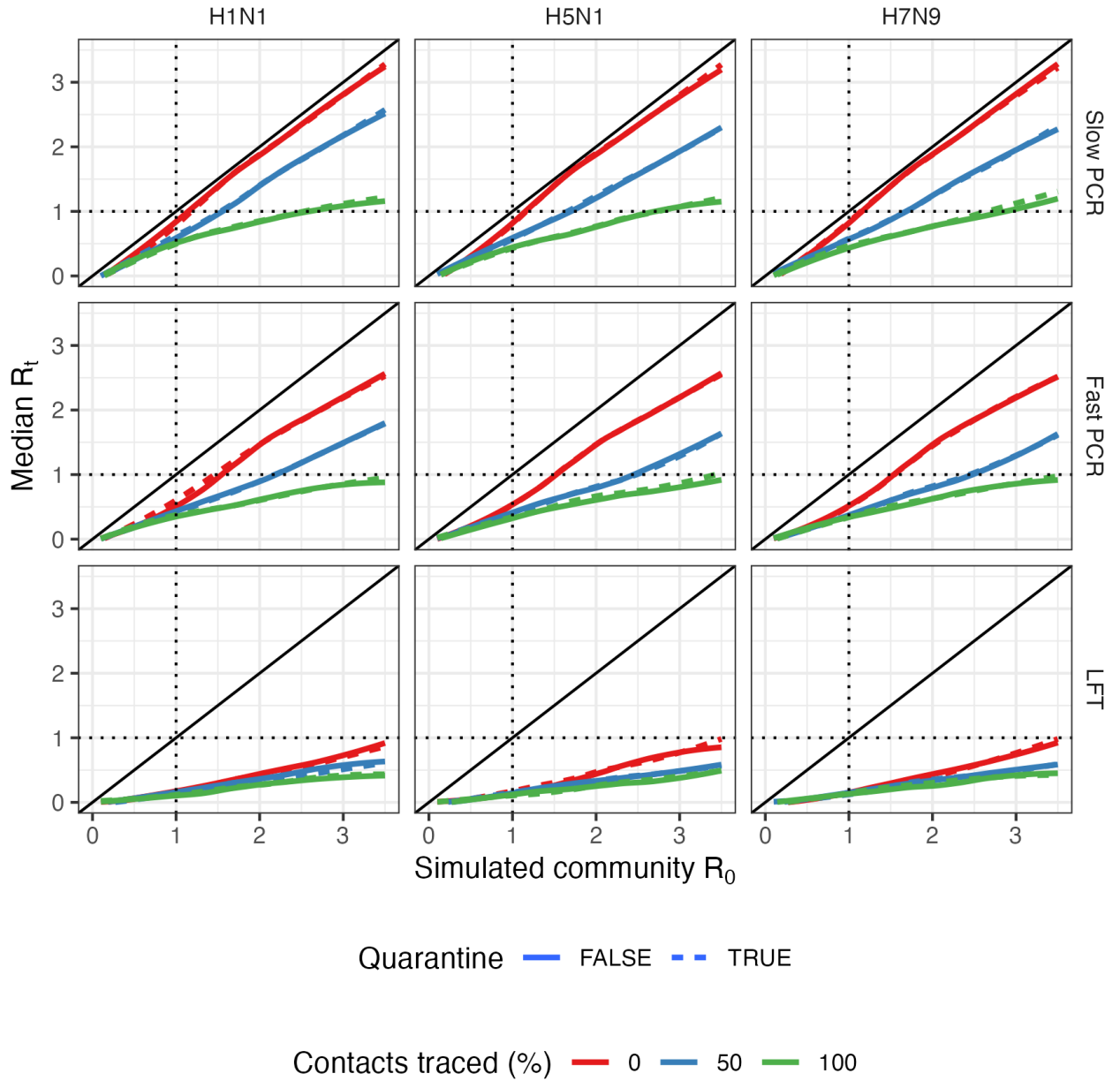


Figure 6: The median effective reproduction number (R_t) across simulation replicates calculated from simulated outbreak data under various influenza subtype and intervention scenarios against the basic reproduction number (R_0) for the general community used to simulate the outbreaks. The R_0 value for isolated cases is zero. The diagonal line shows an equal value for R_t and community R_0 . The dotted lines are where the values of R_t (horizontal) or community R_0 (vertical) equal unity. Values of R_t below the diagonal indicate a reduction in transmission from NPIs compared to uncontrolled community transmission. Isolation of symptomatic cases is active for all scenarios, contact tracing ascertainment is plotted for 0% (no contact tracing), 50% and 100% (all symptomatic contacts of symptomatic cases traced), we run each scenario with quarantine of contacts active (dashed line) and inactive (solid line). This is for a baseline scenario where the proportion of presymptomatic transmission is 1% and the proportion of asymptomatic cases is 0%.

Discussion

Influenza remains a high risk pathogen with pandemic potential subtypes as shown by the history of influenza pandemics throughout the 20th century (Saunders-Hastings and Krewski, 2016). The recent spread of A/H5N1 avian influenza in cattle farms in the US, and poultry farms in the UK and US among others, creates a greater spillover risk and adaptations to enhance mammal-to-mammal transmission (Lin et al., 2024; Garg et al., 2025; Peacock et al., 2025). In this study we use a mathematical model to assess the feasibility and effectiveness of isolation and contact tracing for avian and non-avian influenza subtypes to determine whether these should be considered effective NPIs in the event that sustained human-to-human transmission is detected.

We find that the percentage of contacts that need to be successfully traced to control an outbreak depends on the reproduction number of the pathogen, and the speed at which a public health system can respond by testing and communicating with those infected and contacts of infectious persons. If the delay between symptom onset and isolation is on average 5 days, isolation and comprehensive contact tracing are required to be confident a moderately transmissible ($R_0 \leq 1.5$) influenza can be controlled. By reducing the delay between symptom onset and isolation to around three days, influenza pathogens that are marginally supercritical ($R_0 = 1.1$) become subcritical ($R_t < 1$) and contained by isolation alone, and more transmissible scenarios can be controlled with good contact tracing coverage. In addition to PCR-based responses, testing a rapid LFT-based isolation scenario not requiring tests to be sent to a laboratory, shows feasible opportunities to control highly transmissible ($R_0 \geq 2.5$) respiratory diseases with isolation and contact tracing. The advantage of rapidly testing and isolating symptomatic cases is maintained if the LFT test sensitivity is as good as existing influenza A or SARS-CoV-2 LFTs ($> 80\%$). Therefore, the dogma that symptom-based isolation and contact tracing are not effective for influenza is shown not to hold for optimal testing and isolation scenarios, when leveraging accessible rapid testing, even if some cases are missed due to being asymptomatic and some transmission occurs prior to symptom onset (Fraser et al., 2004; Klinkenberg et al., 2006; Organization, 2017). A pandemic influenza response should focus on reducing the onset-to-isolation delay rather than scaling up a comprehensive contact tracing system. If pathogen transmissibility is high, rapid testing and contact tracing should be deployed. Whether the interventions proposed here are cost-effective remains a question for further investigation.

The transmission model highlights the bimodal nature of the final size of outbreaks, where NPI-controlled epidemics typically range from tens to hundreds of total cases, in contrast, uncontrolled outbreaks can produce thousands of new cases each week. The severity of avian influenza subtypes is uncertain, historical estimates are $\sim 30\% - 55\%$ (Tanner et al., 2015), but recent human infections have lower morbidity and mortality (Garg et al., 2025; Krammer et al., 2025). Consequently, the difference in disease burden between containing an outbreak while still small compared to allowing thousands of cases will be substantial. Even if updated severity estimates are significantly lower, widespread infections could still result in a devastating outcome.

It is possible that a sentinel surveillance system or community testing to detect new cases of pandemic potential influenza (e.g. A/H5N1 or A/H7N9), could miss the primary case(s), whether originating from zoonotic spillover or importation. We show that more cases before an NPI is activated hinders the ability to effectively contain the outbreak. The magnitude of this impact is moderated more by transmissibility than influenza subtype for the three variants we considered. If interventions are activated when case prevalence exceeds 40 cases, then even rapidly isolating over 60% of people with 1 days of becoming symptomatic with a near-perfect contact tracing coverage does not guarantee containment. For comparison, on the date of the first fatality in the UK from COVID-19 there were already over 100 confirmed cases and probable sustained widespread community transmission. A subcritical influenza outbreak could cause transient epidemics, during which, adaptation to enhance human-to-human transmission can cause a critical transition to supercritical transition, at which point more importations exponentially decreases containment without interventions (Whittle, 1955; Farrington and Grant, 1999; Antia et al., 2003; Lloyd-Smith et al., 2005). This study does not model this transition, instead we assume supercritical human-to-human transmission potential for A/H5N1 and A/H7N9 to evaluate NPIs because current estimates for pathogen epidemiology ($R_0 < 1$) do not require interventions for containment. Cases detected with routine surveillance without a known animal source may indicate sustained human-to-human transmission and could be a trigger to commence isolation and contact tracing while the number of initial cases is presumed small.

The simulated scenarios and transmission model have several limitations. Firstly, noting aspects of the scenario exploration. The primary results shown for rapid testing are a best-case scenario assuming 100% test sensitivity. We show that when test sensitivity of LFTs is reduced some of the advantage of rapidly testing is reduced, but that if LFTs for a pandemic influenza are comparable to those for COVID-19 then it is likely still the best response of the three analysed. For this study we focused on symptom-based testing. Regular widespread population testing could detect and isolate presymptomatic and asymptomatic cases, improving chances of outbreak control compared to our results, but this would be a resource intensive response depending on the scale of its implementation. We check for sensitivity to isolation effectiveness and airborne versus droplet transmission. However, our assumed transmissibility for imperfect isolation and droplet transmission for asymptomatic cases could be refined as more data becomes available.

Secondly, on aspects of model assumptions. The model does not include social contact mixing patterns between different age groups or network structure, both of which have shown to influence the effectiveness of NPIs, including isolation and contact tracing (Brooks-Pollock, x2, Juul and Strogatz, 2023). Additionally, we do not accommodate behavioural changes during the outbreak. It is known that once an outbreak starts that people adapt their behaviour forming fewer social contacts and reducing possibilities of disease transmission (van Kerkove et al., 2013). The NPIs modelled in this study also assume complete adherence across all individuals in the population. It has been shown that these exact interventions have almost no effect if adherence is low, even if contact tracing coverage is near 100% or if the speed of isolation response is fast (Davies et al., 2021).

We assessed the benefit of contact tracing across a range of possible contact ascertainment levels. In doing so we model up to 5,000 cases before terminating a simulated outbreak as uncontained, but before reaching the case limit we assume that there is no capacity limit on contact tracing. This is an optimistic assumption; in reality surveillance systems can be patchy and have finite capacity, exacerbated in resource-limited settings (Dhillon and Srikrishna, 2018). The scenarios where a low percentage of contacts were traced may be more representative for these situations. The finding that weekly case incidence can reach thousands of new cases within the acute phase of an influenza outbreak highlights how the exponential dynamics of outbreaks, even in the presence of NPIs, can overburden surveillance systems with the volume of testing and isolation required. Aside from public health surveillance capacity, we also ignore potential socio-political factors that may prevent contacting infected individuals or contacts of infected individuals (Dhillon and Srikrishna, 2018). In this study we assume that contacts are known, avoiding the need to define what constitutes a contact. In reality the threshold for duration and distance of a contact for a directly transmitted respiratory disease will influence the ascertainment and volume of contact tracing required by balancing the error rates of contacts that result in infections (Keeling et al., 2020).

The experience of contact tracing from the COVID-19 pandemic has shown that isolating and quarantining contacts can be expedited with the use of digital tracing. As opposed to symptom-based testing and isolating, a contact can be determined by proximity and duration between mobile devices, and if someone tests positive it can immediately notify all recent contacts (Ferretti et al., 2020). This approach would be complementary to the LFT scenario in this study where available rapid tests could be used when notified of an infectious contact (assuming the time between exposure and testing is longer than the detectable latent period). Decentralising contact tracing to devices most people carry around at all times (e.g. smartphones) would also ease capacity limitations of a centralised contact tracing unit. The optimistic scenarios of traditional testing, tracing and isolating included could in theory be improved and epidemic prevention enhanced by integrating digital technology.

Enacting isolation and contact tracing and other NPIs is expected with the emergence of a novel pathogen due to widespread susceptibility and no pharmaceutical interventions. However, influenza has multiple circulating seasonal subtypes and past pandemics, resulting in a complex mosaic of population immunity for different influenza subtypes, with possible immunological imprinting (Gostic et al., 2016). There is also seasonal vaccination availability in many high- and middle income countries (Goldin et al., 2024), stockpiles of vaccines for avian influenza, as well as licensed effective antivirals (Krammer et al., 2025). We show that asymptomatic influenza infectious will detrimentally influence containment with symptom-based isolation and contact tracing. If an epidemic is highly asymptomatic ($> 25\%$ cases), countries with a stockpile of influenza antivirals, could deploy those prophylactically in response (Hayden, 2001). For example, the UK which stockpiles antivirals for $\sim 50\%$ of the population (Wri). Pandemic response could deploy pharmaceutical and non-pharmaceutical interventions in combination to prevent disease burden by suppressing transmission and severity. The stockpile of vaccines for A/H5N1 that shows cross-reactivity with current 2.3.4.4b strain are another line of defence (Khurana et al., 2024). The speed of isolating contacts compared to vaccinating them in the context of a fast generation time and proportion of presymptomatic transmission, renders a ring vaccination less effective (Kucharski et al., 2016; Whittaker et al., 2024). Overall, the response to an influenza pandemic from an avian or swine subtype would be more multifaceted than the baseline containment results from our model.

Our study shows that isolation and contact tracing are effective NPIs that should be considered when evaluating a robust outbreak response to pandemic potential influenza. If responding to a pathogen where human-to-human transmission is sustained but moderate ($R_0 \leq 1.5$) isolation of symptomatic cases alone will bring the reproduction number below one, and comprehensive tracing and isolation of contacts with a short delay to isolation can control outbreaks with $R_0 \leq 3.5$. Whether the next public health emergency is from avian (A/H5N1 and A/H7N9) or swine (A/H1N1) influenza, the response should be broadly similar. If in the acute phase of the outbreak clinical and epidemiological investigation indicate presymptomatic transmission or asymptomatic infections (Liu et al., 2020), or that multiple imported cases have seeded human-to-human transmission chains (Russell et al., 2021), then increasing contact tracing ascertainment and reducing the onset-to-isolation delay will improve chances of containment. Our results broadly cover to novel emerging pathogens that lack pharmaceutical interventions; extending results to other pandemic pathogen candidates depends on the similarity of their epidemiological characteristics.

References

- Written questions and answers - Written questions, answers and statements - UK Parliament. <https://questions-statements.parliament.uk/written-questions/detail/2021-10-15/57051>.
- Tjandra Y. Aditama, Gina Samaan, Rita Kusriastuti, Ondri Dwi Sampurno, Wilfried Purba, null Misriyah, Hari Santoso, Arie Bratasena, Anas Maruf, Elvieda Sariwati, Vivi Setiawaty, Kathryn Glass, Kamalini Lokuge, Paul M. Kelly, and I. Nyoman Kandun. Avian influenza H5N1 transmission in households, Indonesia. *PloS One*, 7(1):e29971, 2012. ISSN 1932-6203. doi: 10.1371/journal.pone.0029971.
- Sheikh Taslim Ali, Amy Yeung, Songwei Shan, Lin Wang, Huizhi Gao, Zhanwei Du, Xiao-Ke Xu, Peng Wu, Eric H Y Lau, and Benjamin J Cowling. Serial Intervals and Case Isolation Delays for Coronavirus Disease 2019: A Systematic Review and Meta-Analysis. *Clinical Infectious Diseases*, 74(4):685–694, March 2022. ISSN 1058-4838, 1537-6591. doi: 10.1093/cid/ciab491.
- Yehya M. Althobaity, Muhammad H. Alkhudaydi, Edward M. Hill, Robin N. Thompson, and Michael J. Tildesley. The time between symptom onset and various clinical outcomes: A statistical analysis of MERS-CoV patients in Saudi Arabia. *Royal Society Open Science*, 11(11):240094, November 2024. ISSN 2054-5703. doi: 10.1098/rsos.240094.
- Rustom Antia, Roland R. Regoes, Jacob C. Koella, and Carl T. Bergstrom. The role of evolution in the emergence of infectious diseases. *Nature*, 426(6967):658–661, December 2003. ISSN 0028-0836, 1476-4687. doi: 10.1038/nature02104.
- A. Azzalini. A Class of Distributions Which Includes the Normal Ones. *Scandinavian Journal of Statistics*, 12(2):171–178, 1985.
- Tom Britton and Gianpaolo Scalia Tomba. Estimation in emerging epidemics: Biases and remedies. *Journal of The Royal Society Interface*, 16(150):20180670, January 2019. ISSN 1742-5689, 1742-5662. doi: 10.1098/rsif.2018.0670.
- Zongqiu Chen, Hui Liu, Jianyun Lu, Lei Luo, Kuibiao Li, Yufei Liu, Eric H.Y. Lau, Biao Di, Hui Wang, Zhicong Yang, and Xincai Xiao. Asymptomatic, Mild, and Severe Influenza A(H7N9) Virus Infection in Humans, Guangzhou, China. *Emerging Infectious Diseases*, 20(9), September 2014. ISSN 1080-6040, 1080-6059. doi: 10.3201/eid2009.140424.
- Benjamin J. Cowling, Lianmei Jin, Eric H. Y. Lau, Qiaohong Liao, Peng Wu, Hui Jiang, Tim K. Tsang, Jiandong Zheng, Vicky J. Fang, Zhaorui Chang, Michael Y. Ni, Qian Zhang, Dennis K. M. Ip, Jianxing Yu, Yu Li, Liping Wang, Wenxiao Tu, Ling Meng, Joseph T. Wu, Huiming Luo, Qun Li, Yuelong Shu, Zhongjie Li, Zijian Feng, Weizhong Yang, Yu Wang, Gabriel M. Leung, and Hongjie Yu. Comparative epidemiology of human infections with avian influenza A H7N9 and H5N1 viruses in China: A population-based study of laboratory-confirmed cases. *Lancet (London, England)*, 382(9887): 129–137, July 2013. ISSN 1474-547X. doi: 10.1016/S0140-6736(13)61171-X.
- Ranu S Dhillon and Devabhaktuni Srikrishna. When is contact tracing not enough to stop an outbreak? *The Lancet Infectious Diseases*, 18(12):1302–1304, December 2018. ISSN 14733099. doi: 10.1016/S1473-3099(18)30656-X.
- Christl A Donnelly, Azra C Ghani, Gabriel M Leung, Anthony J Hedley, Christophe Fraser, Steven Riley, Laith J Abu-Raddad, Lai-Ming Ho, Thuan-Quoc Thach, Patsy Chau, King-Pan Chan, Tai-Hing Lam, Lai-Yin Tse, Thomas Tsang, Shao-Haei Liu, James Hb Kong, Edith Mc Lau, Neil M Ferguson, and Roy M Anderson. Epidemiological determinants of spread of causal agent of severe acute respiratory syndrome in Hong Kong. *The Lancet*, 361(9371):1761–1766, May 2003. ISSN 01406736. doi: 10.1016/S0140-6736(03)13410-1.
- C. P. Farrington and A. D. Grant. The distribution of time to extinction in subcritical branching processes: Applications to outbreaks of infectious disease. *Journal of Applied Probability*, 36(3):771–779, September 1999. ISSN 0021-9002, 1475-6072. doi: 10.1239/jap/1032374633.
- Neil M. Ferguson, Derek A. T. Cummings, Christophe Fraser, James C. Cajka, Philip C. Cooley, and Donald S. Burke. Strategies for mitigating an influenza pandemic. *Nature*, 442(7101):448–452, July 2006. ISSN 1476-4687. doi: 10.1038/nature04795.
- Luca Ferretti, Chris Wymant, Michelle Kendall, Lele Zhao, Anel Nurtay, Lucie Abeler-Dörner, Michael Parker, David Bon-sall, and Christophe Fraser. Quantifying SARS-CoV-2 transmission suggests epidemic control with digital contact tracing. *Science*, 368(6491), May 2020. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.abb6936.
- Christophe Fraser, Steven Riley, Roy M. Anderson, and Neil M. Ferguson. Factors that make an infectious disease outbreak controllable. *Proceedings of the National Academy of Sciences of the United States of America*, 101(16):6146–6151, April 2004. ISSN 0027-8424. doi: 10.1073/pnas.0307506101.

- Christophe Fraser, Christl A. Donnelly, Simon Cauchemez, William P. Hanage, Maria D. Van Kerkhove, T. Déirdre Hollingsworth, Jamie Griffin, Rebecca F. Baggaley, Helen E. Jenkins, Emily J. Lyons, Thibaut Jombart, Wes R. Hinsley, Nicholas C. Grassly, Francois Balloux, Azra C. Ghani, Neil M. Ferguson, Andrew Rambaut, Oliver G. Pybus, Hugo Lopez-Gatell, Celia M. Alpuche-Aranda, Ietza Bojorquez Chapela, Ethel Palacios Zavala, Dulce Ma Espejo Guevara, Francesco Checchi, Erika Garcia, Stephane Hugonnet, Cathy Roth, and WHO Rapid Pandemic Assessment Collaboration. Pandemic potential of a strain of influenza A (H1N1): Early findings. *Science (New York, N.Y.)*, 324(5934):1557–1561, June 2009. ISSN 1095-9203. doi: 10.1126/science.1176062.
- Shikha Garg, Katie Reinhart, Alexia Couture, Krista Kniss, C. Todd Davis, Marie K. Kirby, Erin L. Murray, Sophie Zhu, Vit Kraushaar, Debra A. Wadford, Cara Drehoff, Allison Kohnen, Mackenzie Owen, Jennifer Morse, Seth Eckel, Jessica Goswitz, George Turabelidze, Steve Krager, Anna Unutzer, Emilio R. Gonzales, Cherissa Abdul Hamid, Sascha Ellington, Alexandra M. Mellis, Alicia Budd, John R. Barnes, Matthew Biggerstaff, Michael A. Jhung, Malia Richmond-Crum, Erin Burns, Tom T. Shimabukuro, Timothy M. Uyeki, Vivien G. Dugan, Carrie Reed, and Sonja J. Olsen. Highly Pathogenic Avian Influenza A(H5N1) Virus Infections in Humans. *New England Journal of Medicine*, 392(9):843–854, February 2025. ISSN 0028-4793, 1533-4406. doi: 10.1056/NEJMoa2414610.
- Shoshanna Goldin, Donald Brooks, Pernille Jorgensen, Pushpa Wijesinghe, Heeyoun Cho, Rania Attia, Reena Doshi, Francisco Nogareda, Belinda Herring, Laure Dumolard, Randie Gibson, Christopher Chadwick, Shalini Desai, Alba Vilajeliu, Ann Lindstrand, Stefano Tempia, Joshua Mott, and Sarah Hess. Seasonal influenza vaccination: A global review of national policies in 194 WHO member states in 2022. *Vaccine*, 42(26):126274, December 2024. ISSN 0264410X. doi: 10.1016/j.vaccine.2024.126274.
- Katelyn M. Gostic, Monique Ambrose, Michael Worobey, and James O. Lloyd-Smith. Potent protection against H5N1 and H7N9 influenza via childhood hemagglutinin imprinting. *Science*, 354(6313):722–726, November 2016. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.aag1322.
- Nina Haug, Lukas Geyrhofer, Alessandro Londei, Elma Dervic, Amélie Desvars-Larrive, Vittorio Loreto, Beate Pinior, Stefan Thurner, and Peter Klimek. Ranking the effectiveness of worldwide COVID-19 government interventions. *Nature Human Behaviour*, 4(12):1303–1312, November 2020. ISSN 2397-3374. doi: 10.1038/s41562-020-01009-0.
- F. G. Hayden. Perspectives on antiviral use during pandemic influenza. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 356(1416):1877–1884, December 2001. ISSN 0962-8436. doi: 10.1098/rstb.2001.1007.
- Daihai He, Shi Zhao, Yingke Li, Peihua Cao, Daozhou Gao, Yijun Lou, and Lin Yang. Comparing COVID-19 and the 1918–19 influenza pandemics in the United Kingdom. *International Journal of Infectious Diseases*, 98:67–70, September 2020. ISSN 12019712. doi: 10.1016/j.ijid.2020.06.075.
- Joel Hellewell, Sam Abbott, Amy Gimma, Nikos I Bosse, Christopher I Jarvis, Timothy W Russell, James D Munday, Adam J Kucharski, W John Edmunds, Sebastian Funk, Rosalind M Eggo, Fiona Sun, Stefan Flasche, Billy J Quilty, Nicholas Davies, Yang Liu, Samuel Clifford, Petra Klepac, Mark Jit, Charlie Diamond, Hamish Gibbs, and Kevin Van Zandvoort. Feasibility of controlling COVID-19 outbreaks by isolation of cases and contacts. *The Lancet Global Health*, 8(4):e488–e496, April 2020. ISSN 2214109X. doi: 10.1016/S2214-109X(20)30074-7.
- Joel Hellewell, Sam Abbott, Amy Gimma, Tim Lucas, Sebastian Funk, and Adam Kucharski. *Ringbp: Simulate and Evaluate Contact Tracing Scenarios*, 2025.
- Niall P. A. S. Johnson and Juergen Mueller. Updating the Accounts: Global Mortality of the 1918-1920 "Spanish" Influenza Pandemic. *Bulletin of the History of Medicine*, 76(1):105–115, March 2002. ISSN 1086-3176. doi: 10.1353/bhm.2002.0022.
- I Nyoman Kandun, Erna Tresnaningsih, Wilfried H Purba, Vernon Lee, Gina Samaan, Syahrial Harun, Eka Soni, Chita Setiawati, Tetty Setiawati, Elvieda Sariwati, and Toni Wandra. Factors associated with case fatality of human H5N1 virus infections in Indonesia: A case series. *The Lancet*, 372(9640):744–749, August 2008. ISSN 01406736. doi: 10.1016/S0140-6736(08)61125-3.
- Matt J Keeling, T Deirdre Hollingsworth, and Jonathan M Read. Efficacy of contact tracing for the containment of the 2019 novel coronavirus (COVID-19). *Journal of Epidemiology and Community Health*, 74(10):861–866, October 2020. ISSN 0143-005X, 1470-2738. doi: 10.1136/jech-2020-214051.
- Surender Khurana, Lisa R. King, Jody Manischewitz, Olivia Posadas, Ashish K. Mishra, Dongxiao Liu, John H. Beigel, Rino Rappuoli, John S. Tsang, and Hana Golding. Licensed H5N1 vaccines generate cross-neutralizing antibodies against highly pathogenic H5N1 clade 2.3.4.4b influenza virus. *Nature Medicine*, 30(10):2771–2776, October 2024. ISSN 1078-8956, 1546-170X. doi: 10.1038/s41591-024-03189-y.
- Eu Suk Kim, Pyoeng Gyun Choe, Wan Beom Park, Hong Sang Oh, Eun Jung Kim, Eun Young Nam, Sun Hee Na, Moonsuk Kim, Kyoung-Ho Song, Ji Hwan Bang, Sang Won Park, Hong Bin Kim, Nam Joong Kim, and Myoung-don Oh. Clinical Progression and Cytokine Profiles of Middle East Respiratory Syndrome Coronavirus Infection. *Journal of Korean Medical Science*, 31(11):1717, 2016. ISSN 1011-8934, 1598-6357. doi: 10.3346/jkms.2016.31.11.1717.

- Don Klinkenberg, Christophe Fraser, and Hans Heesterbeek. The Effectiveness of Contact Tracing in Emerging Epidemics. *PLoS ONE*, 1(1):e12, December 2006. ISSN 1932-6203. doi: 10.1371/journal.pone.0000012.
- Florian Krammer, Enikő Hermann, and Angela L. Rasmussen. Highly pathogenic avian influenza H5N1: History, current situation, and outlook. *Journal of Virology*, 99(4):e02209–24, April 2025. ISSN 0022-538X, 1098-5514. doi: 10.1128/jvi.02209-24.
- Adam J. Kucharski. Controlling minor outbreaks is necessary to prepare for major pandemics. *PLOS Biology*, 22(12): e3002945, December 2024. ISSN 1545-7885. doi: 10.1371/journal.pbio.3002945.
- Adam J. Kucharski, Rosalind M. Eggo, Conall H. Watson, Anton Camacho, Sebastian Funk, and W. John Edmunds. Effectiveness of Ring Vaccination as Control Strategy for Ebola Virus Disease. *Emerging Infectious Diseases*, 22(1):105–108, January 2016. ISSN 1080-6040, 1080-6059. doi: 10.3201/eid2201.151410.
- Adam J Kucharski, Timothy W Russell, Charlie Diamond, Yang Liu, John Edmunds, Sebastian Funk, Rosalind M Eggo, Fiona Sun, Mark Jit, James D Munday, Nicholas Davies, Amy Gimma, Kevin Van Zandvoort, Hamish Gibbs, Joel Hellewell, Christopher I Jarvis, Sam Clifford, Billy J Quilty, Nikos I Bosse, Sam Abbott, Petra Klepac, and Stefan Flasche. Early dynamics of transmission and control of COVID-19: A mathematical modelling study. *The Lancet Infectious Diseases*, 20(5):553–558, May 2020. ISSN 14733099. doi: 10.1016/S1473-3099(20)30144-4.
- Sonja Lehtinen, Peter Ashcroft, and Sebastian Bonhoeffer. On the relationship between serial interval, infectiousness profile and generation time. *Journal of The Royal Society Interface*, 18(174):20200756, January 2021. ISSN 1742-5689, 1742-5662. doi: 10.1098/rsif.2020.0756.
- Justin Lessler, Nicholas G. Reich, Derek A. T. Cummings, New York City Department of Health and Mental Hygiene Swine Influenza Investigation Team, H. P. Nair, H. T. Jordan, and N. Thompson. Outbreak of 2009 pandemic influenza A (H1N1) at a New York City school. *The New England Journal of Medicine*, 361(27):2628–2636, December 2009. ISSN 1533-4406. doi: 10.1056/NEJMoa0906089.
- Y. H. Leung, M. P. Li, and S. K. Chuang. A school outbreak of pandemic (H1N1) 2009 infection: Assessment of secondary household transmission and the protective role of oseltamivir. *Epidemiology and Infection*, 139(1):41–44, January 2011. ISSN 0950-2688, 1469-4409. doi: 10.1017/S0950268810001445.
- Chengjun Li and Hualan Chen. H7N9 Influenza Virus in China. *Cold Spring Harbor Perspectives in Medicine*, 11(8):a038349, August 2021. ISSN 2157-1422. doi: 10.1101/cshperspect.a038349.
- Qun Li, Lei Zhou, Minghao Zhou, Zhiping Chen, Furong Li, Huanyu Wu, Nijuan Xiang, Enfu Chen, Fenyang Tang, Dayan Wang, Ling Meng, Zhiheng Hong, Wenxiao Tu, Yang Cao, Leilei Li, Fan Ding, Bo Liu, Mei Wang, Rongheng Xie, Rongbao Gao, Xiaodan Li, Tian Bai, Shumei Zou, Jun He, Jiayu Hu, Yangting Xu, Chengliang Chai, Shiwen Wang, Yongjun Gao, Lianmei Jin, Yanping Zhang, Huiming Luo, Hongjie Yu, Jianfeng He, Qi Li, Xianjun Wang, Lidong Gao, Xinghuo Pang, Guohua Liu, Yansheng Yan, Hui Yuan, Yuelong Shu, Weizhong Yang, Yu Wang, Fan Wu, Timothy M. Uyeki, and Zijian Feng. Epidemiology of Human Infections with Avian Influenza A(H7N9) Virus in China. *New England Journal of Medicine*, 370(6):520–532, February 2014. ISSN 0028-4793, 1533-4406. doi: 10.1056/NEJMoa1304617.
- Qun Li, Xuhua Guan, Peng Wu, Xiaoye Wang, Lei Zhou, Yeqing Tong, Ruiqi Ren, Kathy S.M. Leung, Eric H.Y. Lau, Jessica Y. Wong, Xuesen Xing, Nijuan Xiang, Yang Wu, Chao Li, Qi Chen, Dan Li, Tian Liu, Jing Zhao, Man Liu, Wenxiao Tu, Chuding Chen, Lianmei Jin, Rui Yang, Qi Wang, Suhua Zhou, Rui Wang, Hui Liu, Yinbo Luo, Yuan Liu, Ge Shao, Huan Li, Zhongfa Tao, Yang Yang, Zhiqiang Deng, Boxi Liu, Zhitao Ma, Yanping Zhang, Guoqing Shi, Tommy T.Y. Lam, Joseph T. Wu, George F. Gao, Benjamin J. Cowling, Bo Yang, Gabriel M. Leung, and Zijian Feng. Early Transmission Dynamics in Wuhan, China, of Novel Coronavirus–Infected Pneumonia. *New England Journal of Medicine*, 382(13):1199–1207, March 2020. ISSN 0028-4793, 1533-4406. doi: 10.1056/NEJMoa2001316.
- Ting-Hui Lin, Xueyong Zhu, Shengyang Wang, Ding Zhang, Ryan McBride, Wenli Yu, Simeon Babarinde, James C. Paulson, and Ian A. Wilson. A single mutation in bovine influenza H5N1 hemagglutinin switches specificity to human receptors. *Science*, 386(6726):1128–1134, December 2024. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.adt0180.
- Natalie M. Linton, Andrei R. Akhmetzhanov, and Hiroshi Nishiura. Correlation between times to SARS-CoV-2 symptom onset and secondary transmission undermines epidemic control efforts. *Epidemics*, 41:100655, December 2022. ISSN 17554365. doi: 10.1016/j.epidem.2022.100655.
- Yang Liu, Centre for Mathematical Modelling of Infectious Diseases nCoV Working Group, Sebastian Funk, and Stefan Flasche. The contribution of pre-symptomatic infection to the transmission dynamics of COVID-2019. *Wellcome Open Research*, 5:58, April 2020. ISSN 2398-502X. doi: 10.12688/wellcomeopenres.15788.1.
- J. O. Lloyd-Smith, S. J. Schreiber, P. E. Kopp, and W. M. Getz. Superspreading and the effect of individual variation on disease emergence. *Nature*, 438(7066):355–359, November 2005. ISSN 0028-0836, 1476-4687. doi: 10.1038/nature04153.

- Ira M. Longini, Azhar Nizam, Shufu Xu, Kumnuan Ungchusak, Wanna Hanshaoworakul, Derek A. T. Cummings, and M. Elizabeth Halloran. Containing Pandemic Influenza at the Source. *Science*, 309(5737):1083–1087, August 2005. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.1115717.
- Joanna Merckx, Rehab Wali, Ian Schiller, Chelsea Caya, Genevieve C. Gore, Caroline Chartrand, Nandini Dendukuri, and Jesse Papenburg. Diagnostic Accuracy of Novel and Traditional Rapid Tests for Influenza Infection Compared With Reverse Transcriptase Polymerase Chain Reaction: A Systematic Review and Meta-analysis. *Annals of Internal Medicine*, 167(6):394–409, September 2017. ISSN 0003-4819, 1539-3704. doi: 10.7326/M17-0848.
- Martha P Montgomery, Sinead E Morris, Melissa A Rolfes, Wanitchaya Kittikraisak, Aaron M Samuels, Matthew Biggerstaff, William W Davis, Carrie Reed, and Sonja J Olsen. The role of asymptomatic infections in influenza transmission: What do we really know. *The Lancet Infectious Diseases*, 24(6):e394–e404, June 2024. ISSN 14733099. doi: 10.1016/S1473-3099(23)00619-9.
- Hiroshi Nishiura and Hisashi Inaba. Estimation of the incubation period of influenza A (H1N1-2009) among imported cases: Addressing censoring using outbreak data at the origin of importation. *Journal of Theoretical Biology*, 272(1):123–130, March 2011. ISSN 1095-8541. doi: 10.1016/j.jtbi.2010.12.017.
- World Health Organization. Update: WHO-confirmed human cases of avian influenza A(H5N1) infection, 25 November 2003–24 November 2006. <https://www.who.int/publications/i/item/who-wer-8206-41-47>.
- World Health Organization. Pandemic Influenza Risk Management: A WHO guide to inform & harmonize national and international pandemic preparedness and response. Technical report, World Health Organization, Geneva, 2017.
- Thomas P. Peacock, Louise Moncla, Gytis Dudas, David VanInsberghe, Ksenia Sukhova, James O. Lloyd-Smith, Michael Worobey, Anice C. Lowen, and Martha I. Nelson. The global H5N1 influenza panzootic in mammals. *Nature*, 637(8045):304–313, January 2025. ISSN 0028-0836, 1476-4687. doi: 10.1038/s41586-024-08054-z.
- Tim Peto, Dominic Affron, Babak Afrough, Anita Agasu, Mark Ainsworth, Alison Allanson, Katherine Allen, Collette Allen, Lorraine Archer, Natasha Ashbridge, Iman Aurfan, Miriam Avery, Ellena Badenoch, Priya Bagga, Rishab Balaji, Ella Baldwin, Sophie Barraclough, Carol Beane, John Bell, Tracy Benford, Susan Bird, Marina Bishop, Angela Bloss, Richard Body, Rosie Boulton, Abbie Bown, Carla Bratten, Chris Bridgeman, Dominic Britton, Tim Brooks, Margaret Broughton-Smith, Pauline Brown, Beverley Buck, Elaine Butcher, Wendy Byrne, Gloria Calderon, Siobhan Campbell, Olivia Carr, Penny Carter, Daniel Carter, Megan Cathrall, Matthew Catton, Jim Chadwick, David Chapman, Kevin K. Chau, Tanzina Chaudary, Shaolin Chidavaenzi, Samatha Chilcott, Bea Choi, Hannah Claasen, Simon Clark, Richard Clarke, Dawn Clarke, Richard Clayton, Kayleigh Collins, Rima Colston, James Connolly, Eloise Cook, Marie Corcoran, Ben Corley, Laura Costello, Caroline Coulson, Ant Crook, Derrick W. Crook, Silvia D’Arcangelo, Mary-Anne Darby, John Davis, Rosaline De Koning, Pauline Derbyshire, Pam Devall, Mark Dolman, Natalie Draper, Mark Driver, Sarah Dyas, Emily Eaton, Joy Edwards, Ruth Elderfield, Kate Ellis, Graham Ellis, Sue Elwell, Rachel Evans, Becky Evans, Marion Evans, Ranoromanana Evans, David Eyre, Codie Fahey, Vanessa Fenech, Janet Field, Alice Field, Tom Foord, Tom Fowler, Mollie French, Hannah Fuchs, Jasmine Gan, Joseph Gernon, Geeta Ghadiali, Narindar Ghuman, Kerry Gibbons, Gurvinder Gill, Kate Gilmour, Anika Goel, Sally Gordon, Tillie Graham, Alexander Grassam-Rowe, David Green, Anna Gronert, Tegan Gumsley-Read, Claire Hall, Bassam Hallis, Sally Hammond, Peter Hammond, Beth Hanney, Victoria Hardy, Gabriella Harker, Andrew Harris, May Havinden-Williams, Elena Hazell, Joanne Henry, Kim Hicklin, Kelly Hollier, Ben Holloway, Sarah J. Hoosdally, Susan Hopkins, Lucy Hughes, Steve Hurdowar, Sally-Anne Hurford, Joanne Jackman, Harriet Jackson, Ruth Johns, Susan Johnston, Juliet Jones, Tinashe Kanyowa, Katie Keating-Fedders, Sharon Kempson, Iftikhar Khan, Beinn Khulusi, Thomas Knight, Anuradha Krishna, Patrick Lahert, Zoe Lampshire, Daniel Lasserson, Kirsten Lee, Lennard Y.W. Lee, Arabella Legard, Cristina Leggio, Justin Liu, Teresa Lockett, Christopher Logue, Vanessa Lucas, Sheila F. Lumley, Vindhya Maripuri, Des Markham, Emma Marshall, Philippa C. Matthews, Sarah McKee, Deborah F. McKee, Neil McLeod, Antoinette McNulty, Freddie Mellor, Rachel Michel, Alex Mighiu, Julie Miller, Zarina Mirza, Heena Mistry, Jane Mitchell, Mika Erik Moeser, Sophie Moore, Akhila Muthuswamy, Daniel Myers, Gemma Nanson, Mike Newbury, Scott Nicol, Harry Nuttall, Jewel Jones Nwanaforo, Louise Oliver, Wendy Osbourne, Jake Osbourne, Ashley Otter, Jodie Owen, Sulaksan Panchalingam, Dimitris Papoulidis, Juan Dobaldo Pavon, Arro Peace, Karen Pearson, Liam Peck, Ashley Pegg, Suzannah Pegler, Helen Permain, Prem Perumal, Leon Peto, Tim E.A. Peto, Thanh Pham, Hayleah L. Pickford, Mark Pinkerton, Michelle Platton, Ashley Price, Emily Protheroe, Hellen Purnell, Lottie Rawden, Sara Read, Charles Reynard, Susan Ridge, Tom G. Ritter, James Robinson, Patrick Robinson, Gillian Rodger, Cathy Rowe, Bertie Rowell, Alexandra Rowlands, Sarah Sampson, Kathryn Saunders, Rachel Sayers, Jackie Sears, Richard Sedgewick, Laura Seeney, Amanda Selassie, Lloyd Shail, Jane Shallcross, Lucy Sheppard, Anna Sherkat, Shelha Siddiqui, Alex Sienkiewicz, Lavanya Sinha, Jennifer Smith, Ella Smith, Emma Stanton, Thomas Starkey, Aleksander Stawiarski, Amelia Sterry, Joe Stevens, Mark Stockbridge, Nicole Stoesser, Anila Sukumaran, Angela Sweed, Sami Tatar, Hema Thomas, Carly Tibbins, Sian Tiley, Julie Timmins, Cara Tomas-Smith, Oliver Topping, Elena Turek, Toi Neibler, Kate Trigg-Hogarth, Elizabeth Truelove, Chris Turnbull, David Tyrrell, Alison Vaughan, John Vertannes, Richard Vipond, Linda Wagstaff, Joanne Waldron, Philip Walker, Ann Sarah Walker, Mary Walters, Jenny Y Wang, Ellie

- Watson, Kate Webberley, Kimerbley Webster, Grace Westland, Ian Wickens, Jane Willcocks, Herika Willis, Stephen Wilson, Barbara Wilson, Louise Woodhead, Deborah Wright, Bindhu Xavier, Fiona Yelnoorkar, Lisa Zeidan, and Rangenzi Zinyama. COVID-19: Rapid antigen detection for SARS-CoV-2 by lateral flow assay: A national systematic evaluation of sensitivity and specificity for mass-testing. *EClinicalMedicine*, 36:100924, June 2021. ISSN 25895370. doi: 10.1016/j.eclinm.2021.100924.
- Timothy W Russell, Joseph T Wu, Sam Clifford, W John Edmunds, Adam J Kucharski, and Mark Jit. Effect of internationally imported cases on internal spread of COVID-19: A mathematical modelling study. *The Lancet Public Health*, 6(1):e12–e20, January 2021. ISSN 24682667. doi: 10.1016/S2468-2667(20)30263-2.
- Patrick Saunders-Hastings and Daniel Krewski. Reviewing the History of Pandemic Influenza: Understanding Patterns of Emergence and Transmission. *Pathogens*, 5(4):66, December 2016. ISSN 2076-0817. doi: 10.3390/pathogens5040066.
- Mrinank Sharma, Sören Mindermann, Charlie Rogers-Smith, Gavin Leech, Benedict Snodin, Janvi Ahuja, Jonas B. Sandbrink, Joshua Teperowski Monrad, George Altman, Gurpreet Dhaliwal, Lukas Finnveden, Alexander John Norman, Sebastian B. Oehm, Julia Fabienne Sandkühler, Laurence Aitchison, Tomáš Gavenčíak, Thomas Mellan, Jan Kulveit, Leonid Chindelevitch, Seth Flaxman, Yarin Gal, Swapnil Mishra, Samir Bhatt, and Jan Markus Brauner. Understanding the effectiveness of government interventions against the resurgence of COVID-19 in Europe. *Nature Communications*, 12(1): 5820, October 2021. ISSN 2041-1723. doi: 10.1038/s41467-021-26013-4.
- Chorh-Chuan Tan. SARS in Singapore—key lessons from an epidemic. *Annals of the Academy of Medicine, Singapore*, 35(5): 345–349, May 2006. ISSN 0304-4602.
- W. D. Tanner, D. J. A. Toth, and A. V. Gundlapalli. The pandemic potential of avian influenza A(H7N9) virus: A review. *Epidemiology and Infection*, 143(16):3359–3374, December 2015. ISSN 1469-4409. doi: 10.1017/S0950268815001570.
- Raymond Tellier, Yuguo Li, Benjamin J. Cowling, and Julian W. Tang. Recognition of aerosol transmission of infectious agents: A commentary. *BMC Infectious Diseases*, 19(1):101, December 2019. ISSN 1471-2334. doi: 10.1186/s12879-019-3707-y.
- Jack Ward, Joshua W. Lambert, Timothy W. Russell, James M. Azam, Adam J. Kucharski, Sebastian Funk, Billy J. Quilty, Oswaldo Gressani, Niel Hens, and W. John Edmunds. Estimates of epidemiological parameters for H5N1 influenza in humans: A rapid review. *medRxiv : the preprint server for health sciences*, 2024. doi: 10.1101/2024.12.11.24318702.
- Charles Whittaker, Gregory Barnsley, Daniela Olivera Mesa, Daniel J Laydon, Chee Wah Tan, Feng Zhu, Rob Johnson, Patrick Doohan, Gemma Nedjati-Gilani, Peter Winskill, Alexandra B. Hogan, Armindeer Deol, Christinah Mukandavire, Katharina Hauck, David Chien Boon Lye, Lin-Fa Wang, Oliver J. Watson, and Azra C Ghani. Quantifying the impact of a broadly protective sarbecovirus vaccine in a future SARS-X pandemic, August 2024.
- P. Whittle. The Outcome of a Stochastic Epidemic—A Note on Bailey’s Paper. *Biometrika*, 42(1/2):116, June 1955. ISSN 00063444. doi: 10.2307/2333427.
- WHO COVID-19 Dashboard. COVID-19 deaths | WHO COVID-19 dashboard. <https://data.who.int/dashboards/covid19/cases>.
- W. Xu, L. Lu, B. Shen, J. Li, J. Xu, and S. Jiang. Serological Investigation of Subclinical Influenza A(H7H9) Infection Among Healthcare and Non-Healthcare Workers in Zhejiang Province, China. *Clinical Infectious Diseases*, 57(6):919–921, September 2013. ISSN 1058-4838, 1537-6591. doi: 10.1093/cid/cit396.
- Yang Yang, M. Elizabeth Halloran, Jonathan D. Sugimoto, and Ira M. Longini. Detecting human-to-human transmission of avian influenza A (H5N1). *Emerging Infectious Diseases*, 13(9):1348–1353, September 2007. ISSN 1080-6040. doi: 10.3201/eid1309.070111.
- Juanjuan Zhang, Maria Litvinova, Wei Wang, Yan Wang, Xiaowei Deng, Xinghui Chen, Mei Li, Wen Zheng, Lan Yi, Xinhua Chen, Qianhui Wu, Yuxia Liang, Xiling Wang, Juan Yang, Kaiyuan Sun, Ira M Longini, M Elizabeth Halloran, Peng Wu, Benjamin J Cowling, Stefano Merler, Cecile Viboud, Alessandro Vespignani, Marco Ajelli, and Hongjie Yu. Evolving epidemiology and transmission dynamics of coronavirus disease 2019 outside Hubei province, China: A descriptive and modelling study. *The Lancet Infectious Diseases*, 20(7):793–802, July 2020. ISSN 14733099. doi: 10.1016/S1473-3099(20)30230-9.

Supplementary Material

Supplementary Methods

Pandemic potential influenza subtypes

In this study we evaluate the feasibility of controlling outbreaks of selected influenza subtypes that have previously caused sporadic human-to-human outbreaks with pandemic potential. We include A/H5N1, A/H1N1, and A/H7N9. These have caused outbreaks of various size and severity over the past 100 years. H5N1 has caused small outbreaks (< 10 cases) of human-to-human transmission (Yang et al., 2007; Aditama et al., 2012). H1N1 is now an endemic seasonal influenza with substantial population immunity, but a novel strain, H1N1pdm09, emerged in 2009 and caused a pandemic outbreak (Fraser et al., 2009; Lessler et al., 2009). The first human case with the H7N9 subtype was in 2013, subsequently causing five outbreak waves, but no human cases since 2019 (Li and Chen, 2021). Most cases are of zoonotic origin with recorded animal exposure but there have been some small clusters (2 — 3 cases) of possible human-to-human transmission (Li et al., 2014). For the purposes of this study we assume that there is no preexisting population immunity for these influenza subtypes. This is generally true for H5N1 and H7N9 at present, and for H1N1pdm09 prior to 2009.

Transmission model generation time from incubation period

Here we provide a more detailed description of how the generation time is calculated from the incubation period. The time between an infector being infected and then infecting an infectee (i.e. generation time) is computed using the incubation period (which samples the symptom onset time from the exposure time of each case) and sampling from a skew-normal distribution (Azzalini, 1985). This approach enables simulating transmission dynamics with a correlated structure between each case's incubation period and generation time for realistic delays between exposure, onset of symptoms, and onward transmission (Hellewell et al., 2020; Linton et al., 2022). This allows the proportion presymptomatic transmission to be specified, rather than defining incubation period and generation time distributions, which only indirectly allows assumptions of presymptomatic infectiousness to be derived. The strength of our approach is based on the simple parameterisation of the individual-based model to capture the infectiousness, disease progression and onward transmission incorporating individual-level heterogeneity. Sampling the generation time using the incubation period and proportion of presymptomatic infections (using a skew-normal distribution) enables simulating transmission dynamics for novel pathogens and early in outbreaks when the incubation period is estimated but the generation time is unknown or estimated with more uncertainty, and the serial interval would introduce bias if used instead (Britton and Scalia Tomba, 2019; Lehtinen et al., 2021).

Given the incubation period for each case, the proportion of the infections that occur prior to, or after, the symptom onset of an infector is controlled by the parameterisation of the skew normal distribution (Figure 2C). The symptom onset time is used as the location parameter (ξ) of the distribution, providing the point around which the distribution is dispersed. The scale parameter (ω) is fixed at 2, this parameter (with the shape parameter) defines the variance of the distribution. We select this value as it provides a realistic amount of variance when transforming onset time to exposure time (variance range for scenarios in this study: 1.5 — 3.2). The shape parameter (α) defines the skewness of the distribution, whereby positive values cause right-skewed values distribution around the location ξ , and negative values have the opposite effect (as $\alpha \rightarrow \infty$, the skew-normal converges to a folded-normal distribution). In summary, for a given parameterisation of the skew-normal distribution given the time of symptom onset (ξ) and fixing ω and α we sample the time an infectee is exposed/infected by an infector.

We parameterise the skew-normal with three proportion of presymptomatic transmission scenarios: 1%, 15%, 30%, these correspond to α values of 31.82053, 1.962614, and 0.726558, respectively (Figure 2C).

Modelling non-pharmaceutical interventions

The epidemic simulation model uses a single-type branching process with a negative binomial offspring distribution to simulate the spread of a pathogen through a population (infinite population size, without depletion of susceptibility). Influenza subtypes have shown moderate transmission heterogeneity (Fraser et al., 2009; He et al., 2020; Ward et al., 2024), but evidence is limited and the negative binomial allows flexibility between highly heterogeneous and homogeneous individual-level transmission. The mean of the negative binomial offspring distribution is the basic reproduction number (R_0), and this distribution allows the model to be parameterised to capture overdispersion of transmission, in other words, some infectors are superspreaders and infect a disproportionate number of individuals (Lloyd-Smith et al., 2005; Kucharski et al., 2020).

There are three categories for individuals in the simulation, with each having their own parameterisation for the negative binomial offspring distribution: 1) *community* (i.e. general population), 2) *isolated*, and 3) *asymptomatic*. Each group have a distinct R_0 and dispersion (k) parameter due to the assumed differences in transmission dynamics between groups.

The NPI implemented in the transmission model is an isolation of symptomatic cases and contact tracing (Figure 1). When a case is symptomatic they are isolated after a delay. The timing of isolation is drawn from the onset-to-isolation distribution. In the scenario of perfect isolation ($R_0^{\text{iso}} = 0$), any infection times that are sampled to occur after the isolation of an infector are removed. In this scenario where isolation is assumed completely effective at preventing further transmission, it reduces the number of secondary cases if the speed of isolation is quicker than some or all of the secondary infection times. The scenario of imperfect isolation uses a basic reproduction number for isolated individuals of 0.5 ($R_0^{\text{iso}} = 0.5$). Therefore, any infection times after the case is isolated are removed, but a second offspring distribution, parameterised with R_0^{iso} and k^{iso} , is sampled, and any infection times after the isolation time are kept and infection times before isolation are removed. These secondary infections in isolation are added to the secondary infections sampled from the community offspring before becoming isolated. To summarise both the perfect and imperfect isolation scenarios, an infectious individual in the community can become isolated at some point in time (t_{iso}). When the case is in the community it transmits with R_0 and k , but after being isolated transmits with R_0^{iso} and k^{iso} . The overall R_0 is composed of secondary infections before isolation (R_0) and secondary infections after isolation (R_0^{iso}). Symptomatic individuals that are traced are isolated either when 1) the onset-to-isolation delay after symptom onset, or 2) the infector is isolated. The isolation of the infectee is whatever occurs first of these two events.

Onset-to-isolation scenarios

In this study we explore how the speed to isolate cases, or inversely, the time delay from symptom onset to isolation, impacts the ability of control an influenza outbreak. The onset-to-isolation delay can be decomposed into, the delay from noticing symptoms to getting a test, the turnaround time of the test, and entering isolation. We explore three scenarios, two based on laboratory conducted Polymerase Chain Reaction (PCR) testing: fast PCR and slow PCR responses, and one on self-testing with lateral flow tests (LFTs). We modelled the two PCR-based responses as Weibull distributions given that the delay from testing to isolation is strictly positive, and often right-skewed (Zhang et al., 2020). The Weibull distribution is also reported as a good fit for empirical onset-to-isolation delay distribution, but a gamma or lognormal distribution would have provided the same characteristics. The exponential distribution is used to model the LFT-based response because most cases will test soon after developing symptoms and the time between taking the test and getting the result is usually 10-30 minutes (Peto et al., 2021). We assume the time from becoming symptomatic to getting tests, test turnaround time, and entering isolation will all decrease compared to the PCR-based response. We expect the principal factor in longer delays to isolation (i.e. > 1 day) is people waiting to take a test after becoming symptomatic, but that this will decay exponentially. In this study we only consider symptom-triggered testing, so although LFTs could be used in testing strategy that detects cases prior to symptom onset, we do not consider this for the purposes of this study.

The distribution and parameterisation choices for our onset-to-isolation scenarios are not based on a single study, but rather represent the range of response delays reported in the literature, covering various geographical location and pathogen types for past infectious disease outbreaks. In addition to onset-to-isolation delay distributions, we also include onset-to-hospitalisation delay distributions, as for several diseases the patient will be isolated upon hospital admission.

In a systematic review and meta-analysis of COVID-19 onset-to-isolation delays, Ali et al. (2022) found that the pooled mean delay was as fast as 2.4 days (range 2.0–2.7) after the peak of the epidemic, and longer at 6.0 days (range 2.9–12.5) before the epidemic peak. Other coronavirus outbreaks have broadly similar delays to isolation. The SARS outbreak in Singapore and Hong Kong again showed an decrease in onset-to-isolation, from 6.8 days in the second week of the outbreak, to 1.3 days in week 9 in Singapore (Tan, 2006), and from 5.4 days to 3.2 days in Hong Kong (Donnelly et al., 2003); while MERS had a mean delay of 5 days (range 2–11) in South Korea (Kim et al., 2016) and ranged from 4.0 to 4.8 days in Saudi Arabia for different age groups (Althobaity et al., 2024). Influenza responses have shown an onset to first medical care for severe cases of 5 days (range 1–12) for the H7N9 subtype (Chen et al., 2014). A WHO review found the median onset-to-hospitalisation for H5N1 cases between 2003–2006 was 4 days (Organization). Specifically in Indonesia, a H5N1 outbreak had a median onset-to-hospitalisation of 6 days (range 1–16 days) (Kandun et al., 2008). An epidemiological investigation of a school outbreak of H1N1pdm in 2009 found the median delay from onset-to-isolation was 3 days (range 0–7) (Leung et al., 2011).

Defining outbreak control

Outbreak control, or the extinction or elimination of the pathogen, is a binary classification, which is true if there are no new cases from the 12th week of the outbreak onwards, and false otherwise. This is the same definition as used in Hellewell et al. (2020). The 12 week threshold is somewhat arbitrary but is judged to be long enough that the NPIs can bring the effective reproduction number below one for all transmission to cease. The number of generations in an outbreak lasting 12 weeks will be approximately between 18 generations (H5N1 median incubation period and 0.1% presymptomatic transmission) and 84 (H1N1 median incubation period and 30% presymptomatic transmission), although realisations of the stochastic simulation may be greater or less than these estimates from distribution medians.

The maximum number of cases in each simulation replicate before terminating the simulation and concluding it is uncontrolled was set to 5,000. This number should be larger than the expected cluster size from a subcritical outbreak which is given by $N_i/(1 - R_0)$, where N_i is the number of initial cases and R_0 is the basic reproduction number. For example, using the largest initial outbreak size in this study $N_i = 40$, and assuming R_0 is subcritical but very close to unity, $R_0 = 0.99$, gives a possible outbreak size of 4,000 cases. Therefore, our limit that 5,000 cases is an uncontrolled outbreak should not include any outbreaks that will go to extinction eventually but produce transient outbreaks by chance.

For computational efficiency we capped the maximum number of days the simulation could run for to 140 (20 weeks). This is substantially longer than the 12 cutoff evaluating outbreak control so does not influence the results.

Supplementary Results

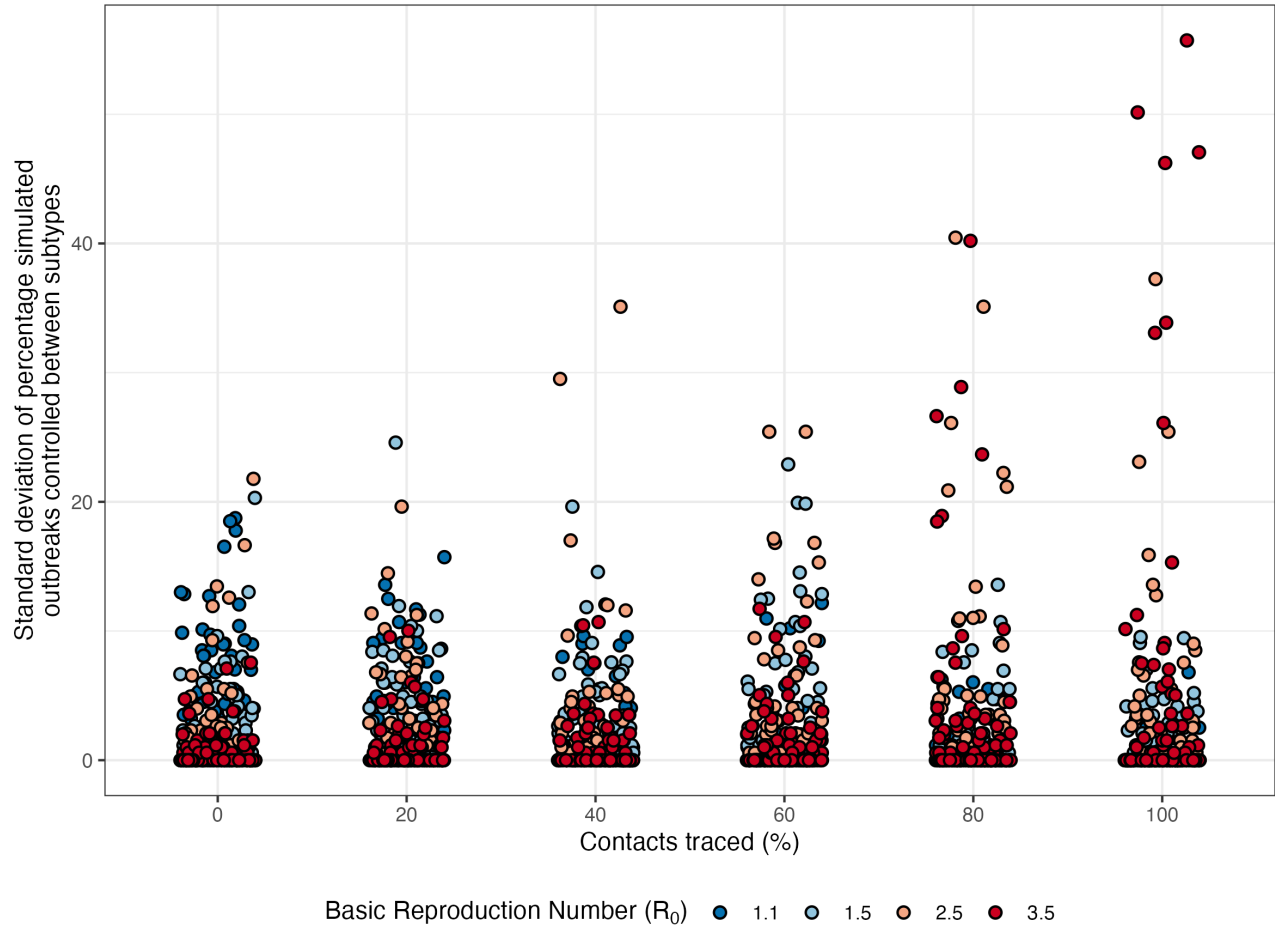


Figure S1: The standard deviation of the percentage of outbreaks controlled between influenza subtypes (H1N1, H5N1, H7N9), across percentages of contacts traced. Points are jittered horizontally reduce overlap.

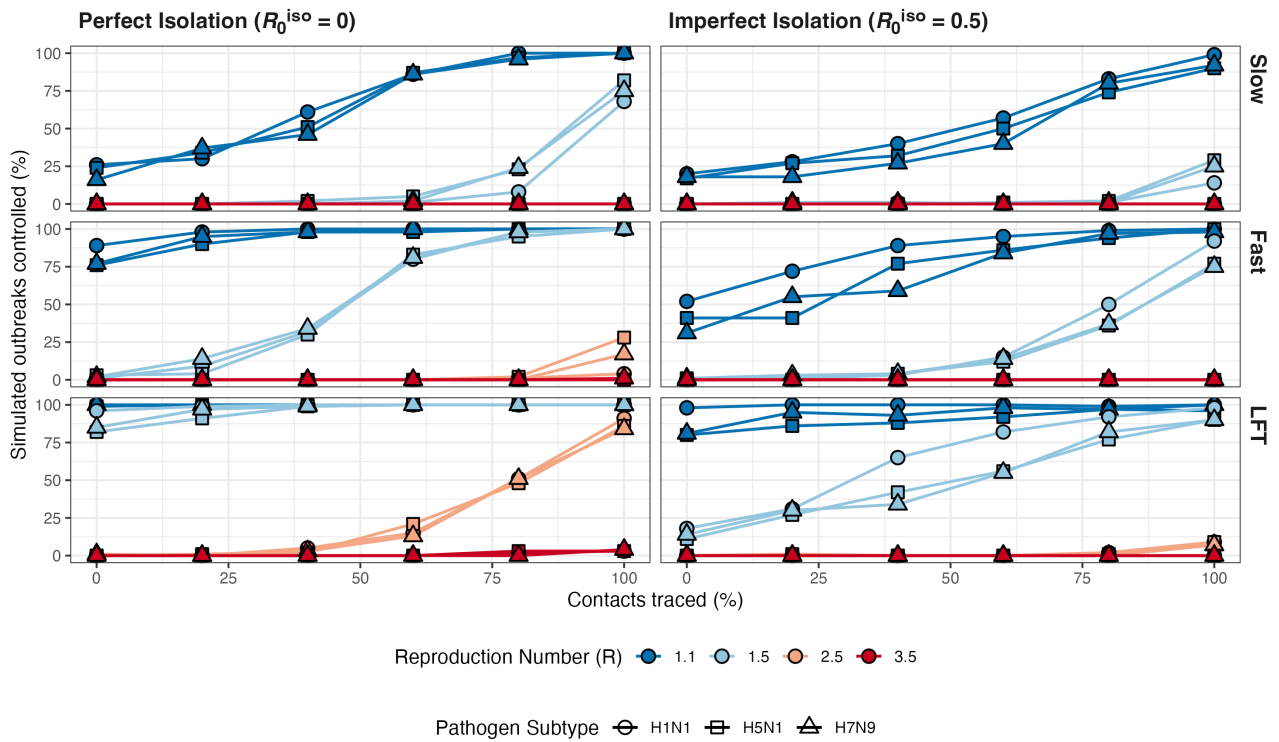


Figure S2: The percentage of outbreaks controlled across values of isolation and contact tracing ascertainment, for outbreaks with an assumed reproduction number (R_0) of either 1.1, 1.5, 2.5 or 3.5. The left column are scenarios with perfect isolation ($R_0^{\text{iso}} = 0$) and the right column are scenarios with imperfect isolation ($R_0^{\text{iso}} = 0.5$). This is for a baseline scenario where the number of initial cases is 20, the proportion of presymptomatic transmission is 15%, 10% are asymptomatic.

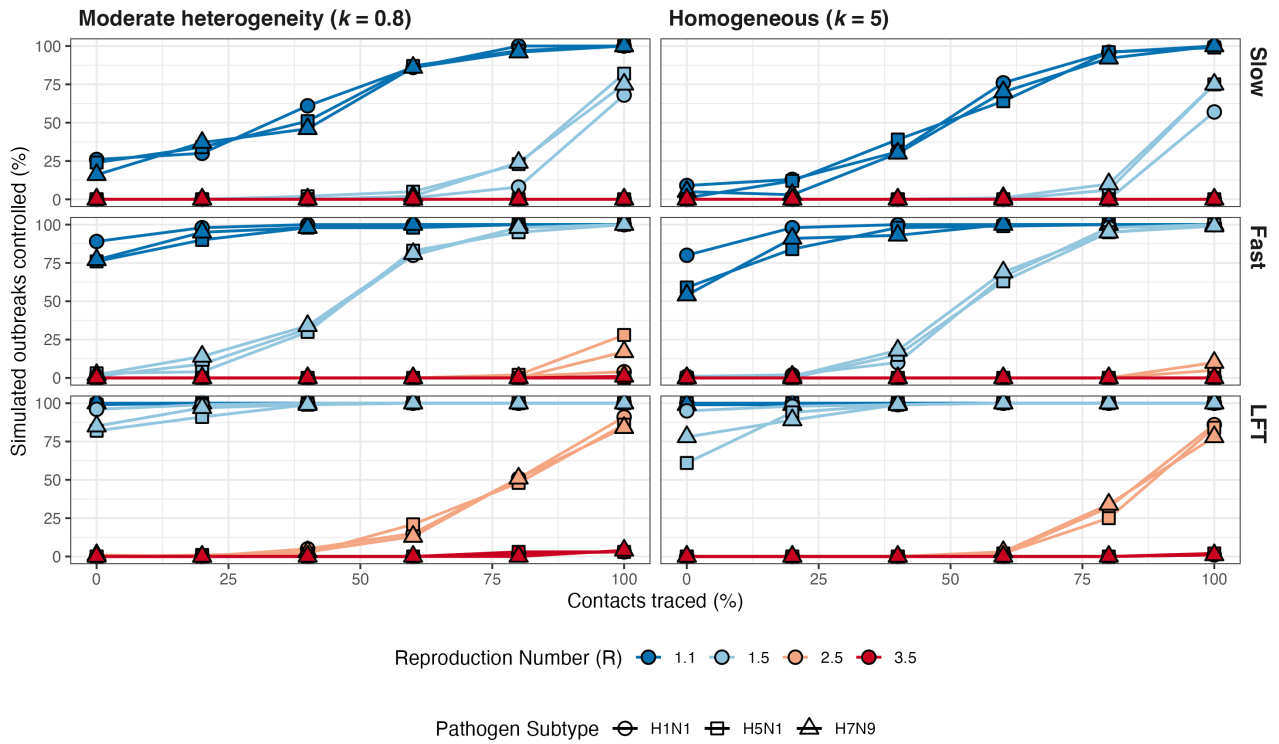


Figure S3: The percentage of outbreaks controlled across values of isolation and contact tracing ascertainment, for outbreaks with an assumed reproduction number (R_0) of either 1.1, 1.5, 2.5 or 3.5. The offspring distribution for secondary cases is modelled as a negative binomial distribution with mean R_0 and dispersion k . The left column are scenarios with moderate individual-level transmission heterogeneity, ($k = 0.8$) and the right column are scenarios with homogeneous transmission ($k = 5$). $k = \infty$ is purely homogeneous, but $k = 5$ is relatively homogenous, and has been shown to be at the lower end for the range of influenza A transmission heterogeneity. This is for a baseline scenario where the number of initial cases is 20, the proportion of presymptomatic transmission is 15%, 10% are asymptomatic.

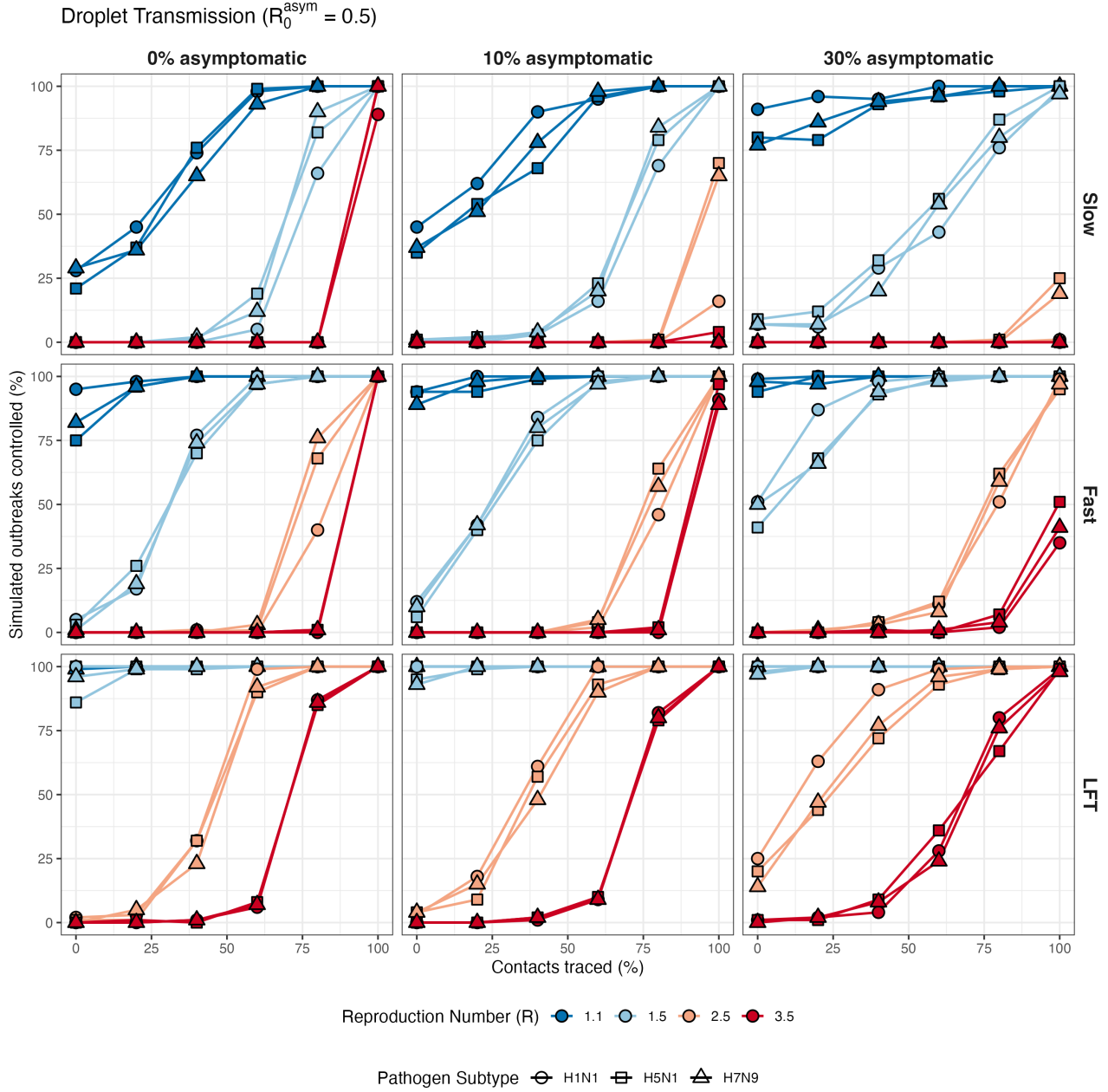


Figure S4: The percentage of outbreaks controlled across values of isolation and contact tracing ascertainment for the droplet transmission scenario. The plot is faceted by the three different proportions of asymptomatic cases simulated (0%, 10% and 30%). The scenarios are plotted by symptomatic community transmissibility ($R_0 = 1.1, 1.5, 2.5, 3.5$) and by pathogen subtype (H1N1, H5N1, H7N9). The droplet transmission scenario assumes transmissibility from asymptomatic cases is reduced relative to symptomatic cases, both of which are in the community ($R_0^{\text{asym}} = 0.5$). The number of initial cases is 20 and the proportion of presymptomatic transmission is 15%.

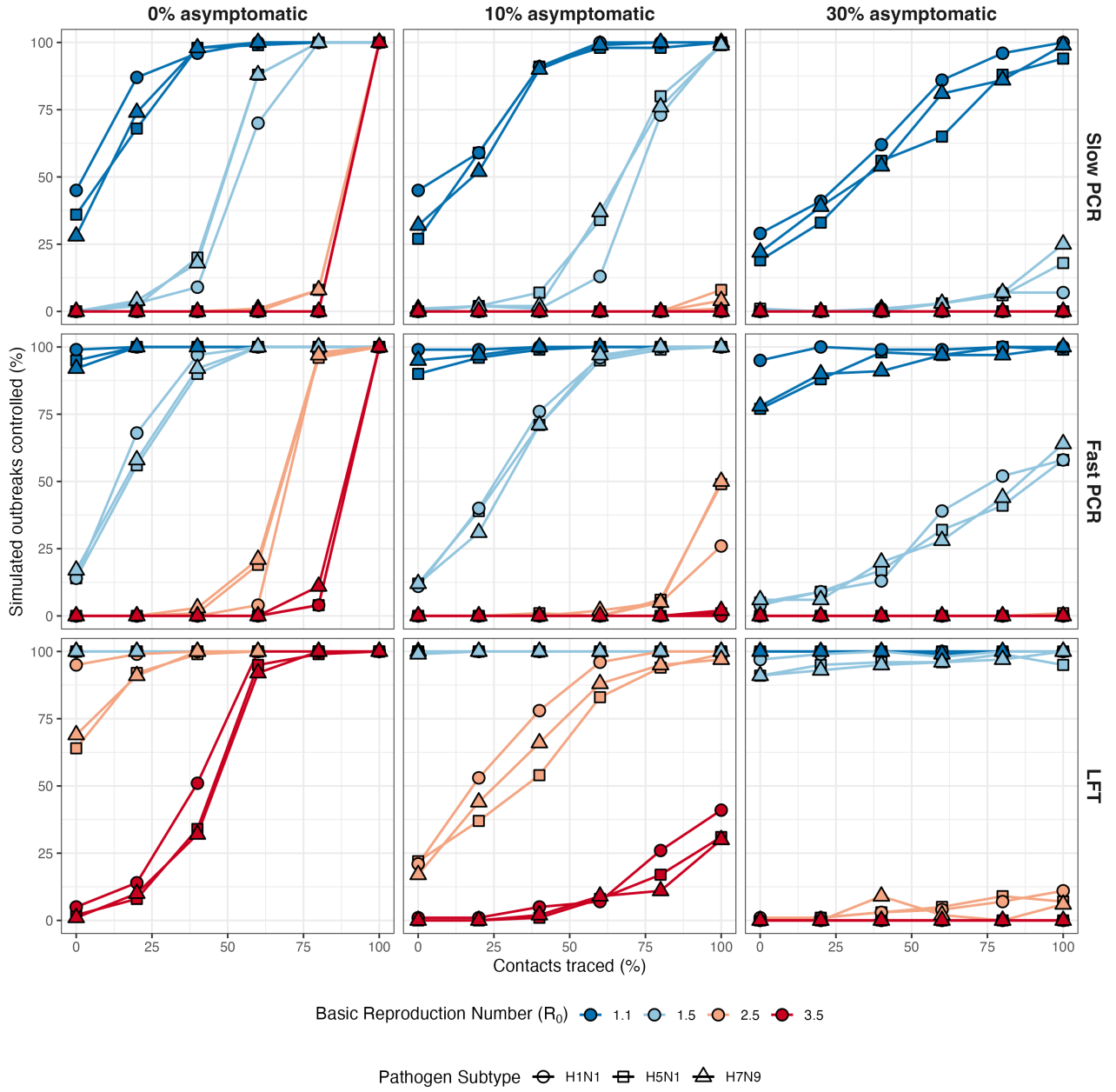


Figure S5: Stub.

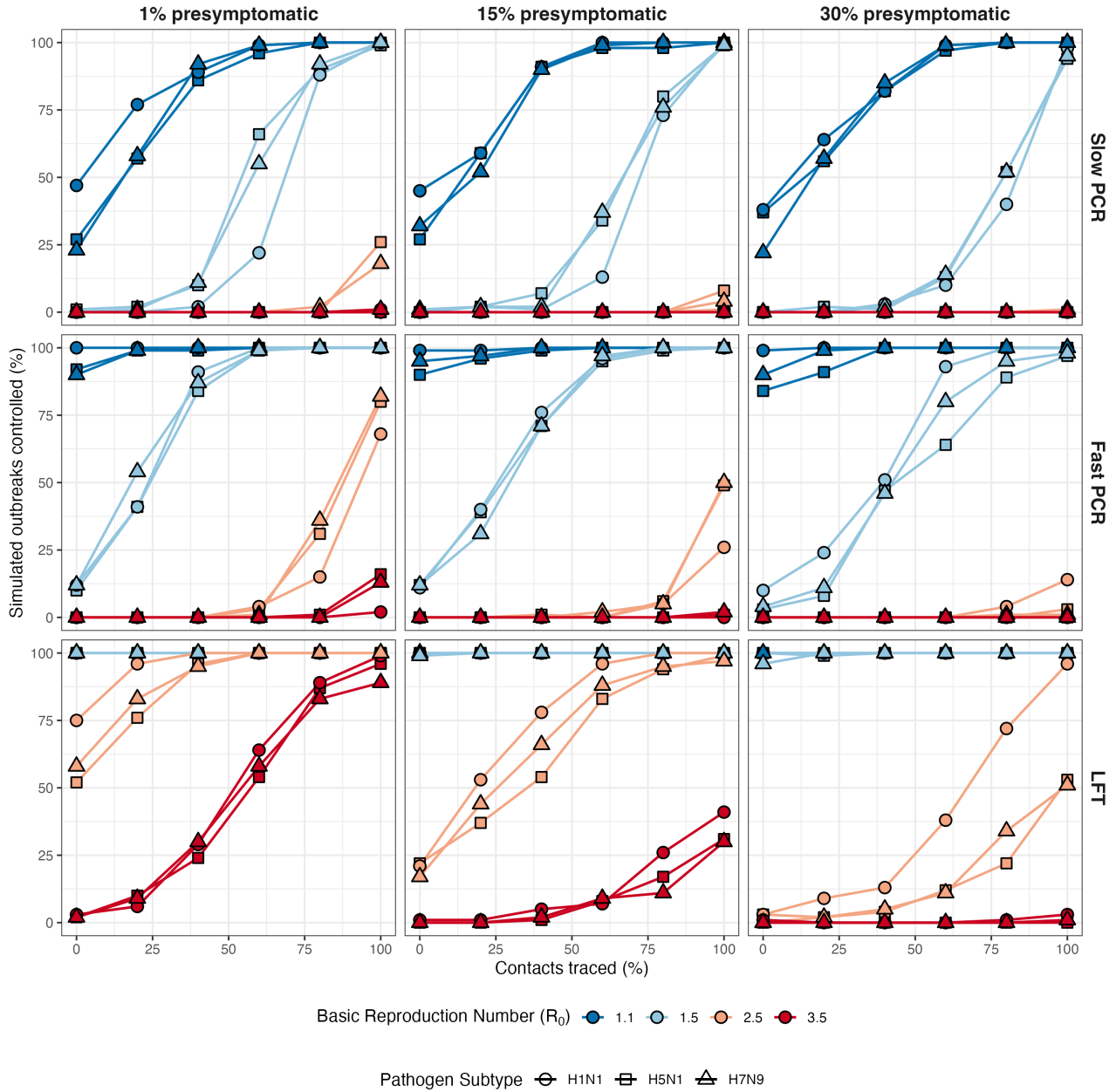


Figure S6: The percentage of outbreaks controlled across percentage contacts traced. Outbreaks vary by basic reproduction number ($R_0 = 1.1, 1.5, 2.5, 3.5$), and influenza subtype (H1N1, H5N1, H7N9). The plot is faceted by percentage of transmission that occurs before symptom onset (presymptomatic), at values of 1%, 15% and 30%, and onset-to-isolation response: slow PCR, fast PCR and LFT. This outbreak simulation had 20 initial cases and 10% of cases are asymptomatic, and contacts are isolated when they become symptomatic (i.e. not quarantined).

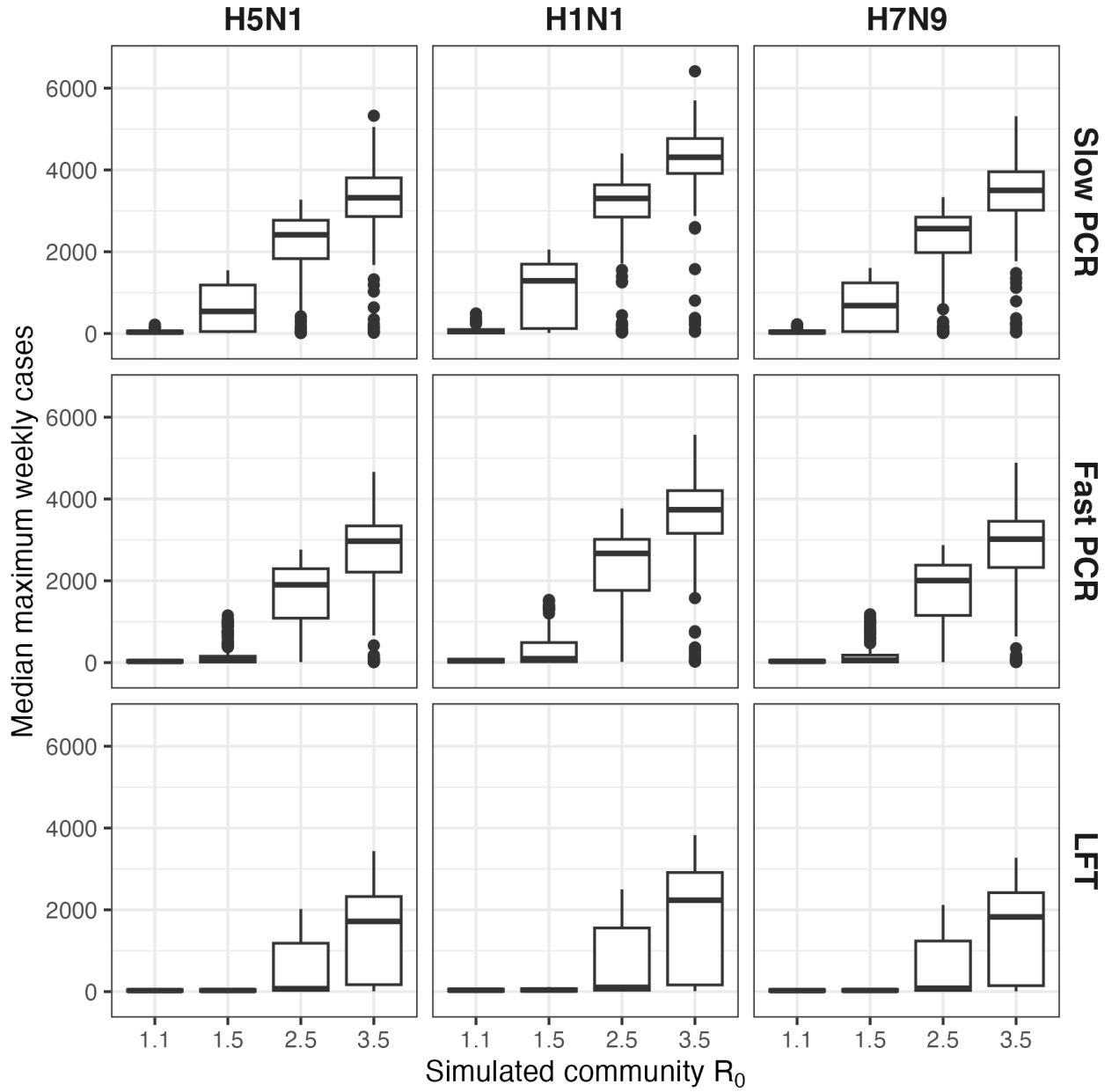


Figure S7: The median maximum number of weekly cases across simulation replicates for different values of community reproduction number (R_0). This is faceted by influenza subtype (A/H5N1, A/H1N1, A/H7N9) and onset-to-isolation response delays (slow COVID-like response, fast SARS-like response and LFT self-testing) The boxplots contain all scenarios (e.g., proportion of asymptomatic cases or number of initial cases) within that reproduction number scenario, the whiskers correspond to the 25th and 75th quartiles Data outside of $1.5 \times$ the interquartile range are plotted as outlier points.

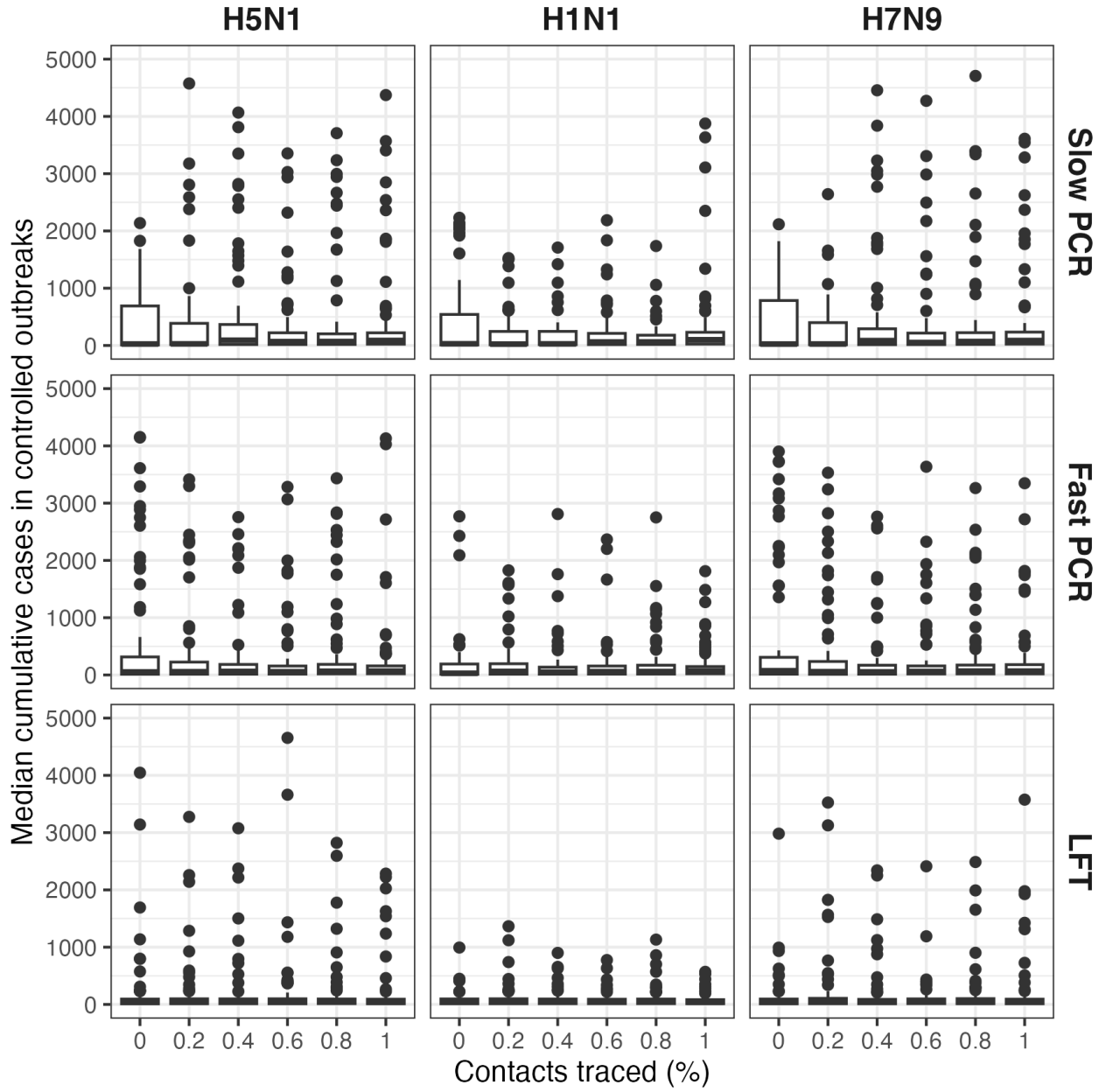


Figure S8: The distribution of the median number of cumulative cases in controlled simulated outbreaks across values of percentage contacts traced. The plot is faceted by influenza subtype: H5N1, H1N1, H7N9, and onset-to-isolation response: slow PCR, fast PCR, and LFT. The boxplot shows where the central tendency and interquartile range for the outbreak size, the whiskers correspond to the 25th and 75th quartiles. Data outside of $1.5 \times$ the interquartile range are plotted as outlier points.

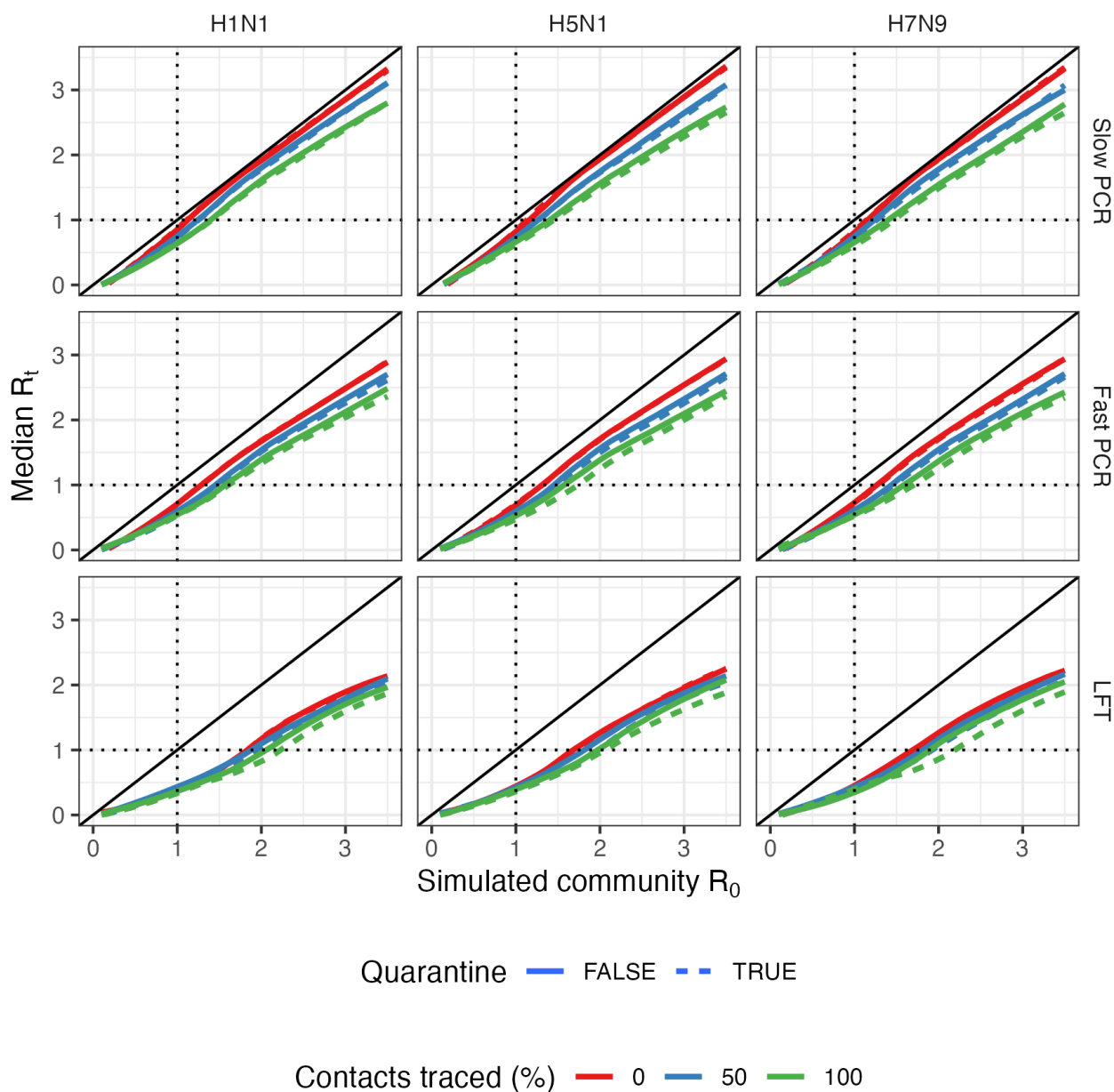


Figure S9: The median effective reproduction number (R_t) across simulation replicates calculated from simulated outbreak data under various influenza subtype and intervention scenarios against the basic reproduction number (R_0) for the general community used to simulate the outbreaks. The R_0 value for isolated cases is zero. The diagonal line shows an equal value for R_t and community R_0 . The dotted lines are where the values of R_t (horizontal) or community R_0 (vertical) equal unity. Values of R_t below the diagonal indicate a reduction in transmission from NPIs compared to uncontrolled community transmission. Isolation of symptomatic cases is active for all scenarios, contact tracing ascertainment is plotted for 0% (no contact tracing), 50% and 100% (all symptomatic contacts of symptomatic cases traced), we run each scenario with quarantine of contacts active (dashed line) and inactive (solid line). This is for a baseline scenario where the proportion of presymptomatic transmission is 1% and the proportion of asymptomatic cases is 0%.